PUNJAB CURRICULUM AND TEXTBOOK BOARD • SNC 2022

Computer Science — Class 7

Easy Exam Notes • Unit 2 to Unit 6

UNITS IN THESE NOTES

2 Digital Skills

3 Computational Thinking

4 Programming

5 Digital Citizenship

6 Entrepreneurship in Digital Age

5 Units • 31 Topics

HOW TO READ EVERY UNIT

1 Topic notes › **2** Exercise Long Question › **3** Full answer › **4** Short Questions › **5** MCQs › **6** Mistakes in the book

- ✔ **117** Short Questions • **175** MCQs • every Exercise Long Question solved
- ✔ Long Questions are taken only from the book exercise and clearly marked
- ✔ Easy English, and every exercise MCQ checked against the book's answer key

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Printed by Mr Books Plus

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Computer Science — Class 7

Single National Curriculum (SNC) — Punjab Curriculum and Textbook Board

Unit 2 — Digital Skills

TOPIC 1: WORD PROCESSOR: A word processor is one of the most used computer programs in schools and offices. It helps us to write, change, save and print text.

Purpose of a word processor (2.1)

A word processor is used to make documents such as letters, notes, reports and homework. It has four main functions:

1. **Composing:** Writing (typing) new text in a document.
2. **Editing:** Making changes in the text, such as adding, removing or correcting words.
3. **Saving:** Storing the document in the computer so that we can open and use it later.
4. **Printing:** Taking a paper copy (hardcopy) of the document from a printer.

Word processing software (2.1.1)

A word processor is a program that allows us to enter, format, save and print text. Text means words and numbers. **Microsoft Word (MS Word)** is a very popular word processing program. It is mainly used to make brochures, letters, learning activities, quizzes, tests and homework assignments.

Examples of word processors: MS Word, Google Docs, WordPerfect, LibreOffice Writer, Dragon, Scrivener and RagTime. Notepad and Open Office are also used to write documents.

Open a new Word document (2.1.2)

When we make a new document, we usually start with a new blank document. We can also start from a ready-made design called a **template**. The steps are:

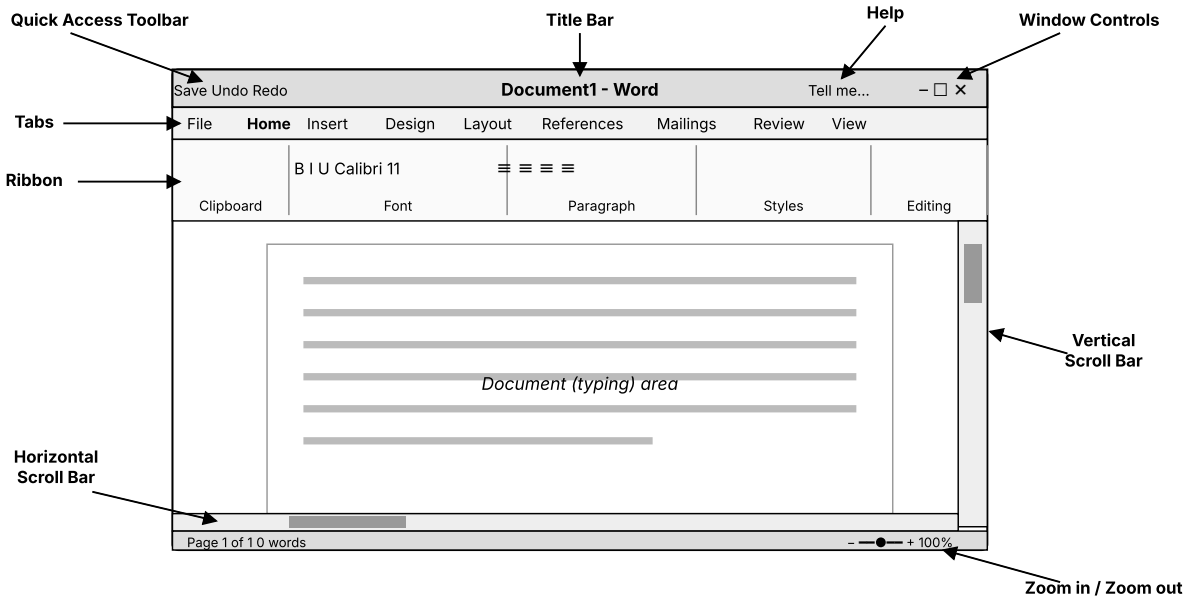
1. Open Word. If Word is already open, select **File > New**.
2. In the "Search for online templates" box, type a word like letter, resume or invoice. Or choose a category like Business, Personal or Education.
3. Click a template to see its preview.
4. Select **Create**. The new document opens.

Interface of MS Word (2.1.3)

The interface is the screen that we see when MS Word is opened. Its main parts are:

1. **Title Bar:** It is at the top. It shows the name of the document, such as "Document1 - Word".

2. **Quick Access Toolbar:** It is on the top-left corner. It has buttons that we use again and again, such as Save, Undo and Redo.
3. **Tabs:** These are names like File, Home, Insert, Design, Layout, References, Mailings, Review and View. Each tab opens its own set of commands.
4. **Ribbon:** It is the wide strip below the tabs. It shows the commands of the selected tab in groups, such as Clipboard, Font, Paragraph and Styles.
5. **Help Button:** It is used to get help about any feature ("Tell me what you want to do").
6. **Window Controls:** These are the Minimize, Maximize and Close buttons on the top-right corner.
7. **Scroll Bars:** The **vertical scroll bar** moves the page up and down. The **horizontal scroll bar** moves the page left and right.
8. **Zoom in / Zoom out:** It is on the bottom-right corner. It makes the page look bigger or smaller on the screen.



Interface of Microsoft Word

Long Question: Describe the purpose of a word processor. ***[Exercise (Brief Q7)]***

1. **Meaning:** A word processor is a computer program that allows us to enter, format, save and print text. Microsoft Word is the most popular word processor.
2. **Composing:** We use it to type new text, such as a letter, an application or our homework.
3. **Editing:** We can easily correct mistakes, add new words or delete wrong words without writing the whole page again.
4. **Formatting:** We can make text bold, italic or underlined. We can also change the font, size, colour and alignment of the text.
5. **Saving:** We can save the document in the computer and open it again later.

6. **Printing:** We can take a paper copy of the document with a printer.

Uses: It is used to make letters, brochures, quizzes, tests, reports and homework assignments.

TOPIC 2: WORKING WITH TEXT IN MS WORD: MS Word has many features that are not found in a simple text editor like a plain-text file. We can change the look of the text, insert pictures and much more.

Adding and changing the text (2.1.4)

The formatting that we apply to text is called **character formatting**. It includes font, font size, bold, italic, underline, strikethrough, subscript, superscript, font colour and highlight colour.

1. **Font Face:** It changes the shape (style of writing) of the letters. It is like changing your handwriting.
2. **Font Size:** It makes the text bigger or smaller.
3. **Font Colour:** It changes the colour of the text.

To format any text, we first **select the text**. Then we use the commands of the **Home tab**.

There are three font styles on the Home tab:

1. **Bold (B):** It makes the text look thick and dark.
2. **Italic (I):** It makes the text slant (lean) towards one side.
3. **Underline (U):** It draws a line under the selected text.

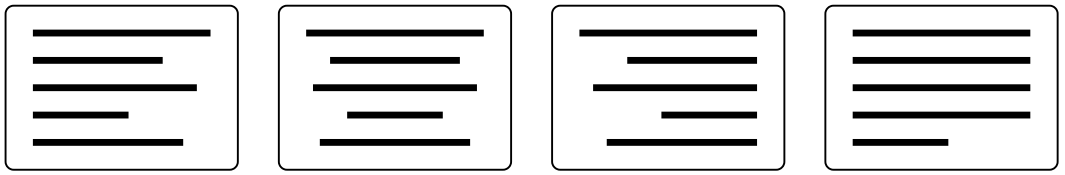
Save and open files (2.1.5)

After writing the text, it is important to save the document. Saving allows us to open and use it later.

1. **To Save:** Go to the **File tab** and press the **Save** option, or click the Save button on the Quick Access Toolbar. Then choose the folder and type the document name.
2. **To Open:** Go to the **File tab** and click the **Open** button. Then select the saved file.

Format a document (2.1.6)

1. **Text Alignment:** Alignment means the position of the text on the page. MS Word has **four** types of alignment: Left Align, Center Align, Right Align and Justify. In Justify, the text touches both the left and the right margins.
2. **Page Margins:** Margins are the blank spaces on the top, bottom, left and right sides of a page. They make a document look neat and professional. To change margins, click the **Margins** button on the **Layout (Page Layout) tab**.
3. **Page Layout:** The **Layout tab** lets us control the look and feel of the whole document. We can apply themes and colour schemes. We can also change the **orientation** (Portrait or Landscape), page size, margins, indentation, line spacing and paragraph settings.



Four types of text alignment

Remember: Portrait orientation means the page stands up (tall). **Landscape** orientation means the page lies on its side (wide).

TOPIC 3: PICTURES, CUT, COPY, PASTE AND SPELL CHECK: MS Word lets us put pictures in a document and move text easily from one place to another.

Insert, resize and add text in images (2.1.7)

To insert an image:

1. Place the cursor where you want to insert the image.
2. Select the **Insert** tab.
3. Select the **Pictures** button in the **Illustrations** group.
4. **Choose where the picture is: This Device** (a picture saved on your computer), **Stock Images** (ready icons, stickers and pictures) or **Online Pictures** (search pictures through Bing; it also gives a link to OneDrive).
5. Find the picture in the Insert Picture dialog box.
6. Click the **Insert** button.

To resize an image: Select the image. A border with small **resizing handles** appears. Drag a handle to make the image bigger or smaller.

To align an image with the text: Select the image, click the **Layout Options** button and choose an option: In Line with Text, Square, Tight, Through, Top and Bottom, Behind Text, or In Front of Text.

Cut, copy and paste (2.1.8)

1. **Cutting:** It removes the text or image from its place and keeps it in the **Clipboard**. The clipboard is a temporary storage area in the computer.
2. **Copying:** It puts a duplicate (copy) of the text or image in the clipboard, but the original stays in its place.
3. **Pasting:** It moves the text or image from the clipboard to a new place in the document.

Keyboard shortcuts

Command	Shortcut	What it does
Cut	Ctrl + X	Removes the selected item and keeps it in the clipboard
Copy	Ctrl + C	Makes a copy of the selected item in the clipboard
Paste	Ctrl + V	Puts the item from the clipboard at the cursor

Ribbon method: Select the text or image, go to the **Home** tab and click Cut or Copy in the **Clipboard** group. This method needs more steps than shortcuts, but it gives more paste options.

Spell check (2.1.9)

Spell check is a built-in tool of MS Word. It checks our writing for spelling and grammar mistakes.

1. If it is turned on, Word automatically underlines the spelling and grammar mistakes while we type.
2. We can also open the **Spelling & Grammar** box from the **Review** tab to check all mistakes at once.
3. The shortcut key for spelling and grammar check is **F7**.

TOPIC 4: LISTS, TABLES, HEADER, FOOTER, THESAURUS AND PRINTING:

These tools help us to arrange our document and make it look professional.

Numbered and bulleted lists (2.1.10)

1. **Numbered List:** Each item has a number or letter (1, 2, 3 or a, b, c). It is used when the items must be in a certain order, like the steps of a method.
2. **Bulleted List:** Each item has a bullet (a black dot or another symbol) but no number. It is used when the order of items does not matter.

Both lists make steps and points simple and easy to read.

To create a bulleted list: Click the **Bullets** button, type the first item and press **Enter**. The next line gets a new bullet automatically. When the list is complete, press **Enter twice** to stop the list.

To create a numbered list: Click the **Numbering** button, type the first item and press **Enter**. The next line gets the next number automatically. Press **Enter twice** to stop the list.

Difference between Save and Save As (2.1.11)

1. **Save:** It updates the last saved file with the new work. It is used to save changes in the same file, with the same name and place. Shortcut: **Ctrl + S** (also Shift + F12 or Alt + Shift + F2).
2. **Save As:** It is used when we save a file for the first time, or when we want to change its name, place (location) or format. It also helps to keep the original file safe as a backup. Shortcut: **F12** (or Alt + F + A).

Save vs Save As

Save	Save As
Stores the data back in the same file and folder	Stores a new file, or the file in a new place
Prevents data loss and updates the file with the latest work	Stores an existing file in a new place, with the same or a different name
Applies to the current file	Can apply to a new file
Has only one step	Needs some extra steps
Does not allow saving in another format	Allows the user to change the file format

Insert and format a table (2.1.12)

1. Place the cursor where you want the table.
2. Select the **Insert** tab in the ribbon.
3. Select the **Table** button in the Tables group.
4. Move the pointer across and down the grid to select the number of rows and columns.
5. The selected cells turn orange. Click the selected part or press **Enter** to insert the table.

Header, footer and page numbers (2.1.13)

Headers and footers are used to add extra information such as the title, file name, date and page numbers. They make a document look more professional and easy to read.

1. **Header:** It appears in the **top margin** of every page.
2. **Footer:** It appears in the **bottom margin** of every page.

Steps: Open the document, go to the **Insert** tab and click **Header**, **Footer** or **Page Number** in the Header & Footer group.

Long Question: Elaborate on the use of thesaurus and synonyms features in Microsoft Word. ***[Exercise (Brief Q5)]***

1. **Thesaurus:** The thesaurus is a tool in MS Word. It is used to find **synonyms** (words with the same meaning) and **antonyms** (words with the opposite meaning) of a selected word.
2. **Use:** It is mostly used when we write an email, a letter, a project report or a research paper. It helps us to choose a better word and to avoid using the same word again and again.
3. Steps to use synonyms:

- Open a new or an existing Word document.
- Type a word, or find a word in the document.
- Highlight (select) the word. A blue background appears behind it.

- Right-click on the selected word. A drop-down menu appears.
- Place the cursor on the **Synonyms** option and click the best word from the list.

We can also open the full **Thesaurus** from the Synonyms menu or from the **Review** tab.

Print a document (2.1.15)

A document can be printed easily if a printer is attached to the computer and its software is installed.

1. Open the Word document by double-clicking it, or by going to **File > Open**.
2. Go to **File > Print**.
3. Choose the printing options (number of copies, printer, pages) and click **Print** or **OK**.

TOPIC 5: MULTIMEDIA PRESENTATION (POWERPOINT): A multimedia presentation is a set of media objects such as pictures, text, video clips and sound, shown on slides one after another.

Meaning of a multimedia presentation (2.2)

In a presentation, every object is shown at a fixed time (in order) and at a fixed place on the screen. The easiest way to make a multimedia presentation is **Microsoft PowerPoint**. We can add video, sound and animation in it. The biggest advantage of PowerPoint is that **anyone can easily edit** the presentation.

Adding new slides (2.2.1)

1. On the **Home** tab, click **New Slide**.
2. In the New Slide box, select the layout you want, such as Title Slide, Title and Content or Section Header.
3. Select **Add Slide**.

Adding objects on slides: image, text, video, audio (2.2.2)

PowerPoint gives many ways to insert clip art, images, audio and graphics. We can import files saved in different formats. PowerPoint also has its own gallery of clip art.

- To insert clip art, choose a slide layout that has a content placeholder and click the **Insert Clip Art** button.

Steps to insert an object:

1. Click on the slide where you want to place the object.
2. On the **Insert** tab, in the Text group, click **Object**.
3. If the object does not already exist, click **Create New**.
4. To show the content on the slide, clear the **Display as icon** check box.

Adding notes (2.2.3)

Speaker notes help the speaker to remember and stress the main points during a presentation. The audience does not see these notes.

1. Open the presentation and click **View > Notes**, or click the **Notes** button at the bottom of PowerPoint.
2. Click on "Click to add notes" and type the notes for that slide. There is no limit to the number of notes.

Creating a slide show and an executable file (2.2.4)

When all slides are ready, press **F5** to start the show from the first slide. Use the **arrow keys** to move between slides.

To set up the slide show, select **Slide Show > Set Up Slide Show** and choose one option:

1. **Browsed by an individual (window):** The show runs inside a window. This works best when the show is saved on a **CD**.
2. **Browsed at a kiosk (full screen):** The show runs automatically on the full screen. This works best when the show is changed into a **video**.

Add timings to each slide: Select **Slide Show > Rehearse Timings**. The show opens on the full screen with a **Recording toolbar** and a timer. Use Next to go to the next slide, Pause to stop the timer and Repeat to record the time again. At the end, close the toolbar and select **Yes** to keep the timings.

Edit timings of a slide: Select **View > Slide Sorter**, select **Transitions**, select a slide and type the time in the **Advance Slide After** box. Then press Enter.

Long Question: Write a note on benefits of the Microsoft PowerPoint which adds importance to any of the presentation for any kind of users. ***[Exercise (Brief Q1)]***

Microsoft PowerPoint is one of the most popular presentation tools. People of all ages use it. It helps us to present our ideas in an attractive way. In business meetings it makes the session more interesting than the old way. Its main benefits are:

1. **Point to Point Focus:** The presentation stays focused on the main points. Instead of long paragraphs, we use short points with pictures, sound or animation. This saves time and effort.
2. **Attractive Visuals:** Pictures stay in our mind longer than spoken words. Colourful backgrounds, text and animation give the slides a striking look.
3. **Numerous Resources:** PowerPoint has many features to add material and references from the Internet with a single click.
4. **Breaking the Complexity:** It helps the presenter to explain difficult ideas in a simple way. A **laser pointer** can be used, so the presenter keeps eye contact with the audience while pointing to the bullet points.
5. **User Ownership:** We can add a **digital signature**. Then nobody can change the presentation without the owner's permission.
6. **Multimedia Capability:** A presentation can be changed into a video or audio file. It can be shared on the Internet or saved on a DVD. It can also be viewed on mobile phones and other devices.
7. **Easy Accessible:** Slides can be easily sent by email or shared through storage devices like a USB.

Long Question: Prepare a presentation on the book the students read last time, while incorporating all the tools. ***[Exercise (Detail Q1)]***

We can prepare a presentation in Microsoft PowerPoint by following these steps:

1. **Open PowerPoint:** Open Microsoft PowerPoint and choose a **Blank Presentation**.
2. **Title Slide:** On the first slide, type the name of the book and your name.
3. **Add New Slides:** Click **Home > New Slide** and choose a layout, such as Title and Content. Add slides for the author, the story, the main characters and the lesson of the book.
4. **Add Text:** Click in the placeholder and type short points about the book.
5. **Add Pictures, Audio and Video:** Click the **Insert** tab and use Pictures, Audio or Video to add the book cover or other media.
6. **Add Speaker Notes:** Click **View > Notes** and write what you will say on each slide.
7. **Add Timings:** Use **Slide Show > Rehearse Timings** to set the time for each slide.
8. **Start the Slide Show:** Press **F5** to show the presentation from the first slide. Use arrow keys to move between slides.
9. **Save:** Press **Ctrl + S** to save the presentation. It can also be saved as a video to share with others.

TOPIC 6: ELECTRONIC MAIL (E-MAIL): Electronic mail, or **email**, is a way of communication that uses electronic devices to send messages across networks.

About email (2.3)

Gmail (Google mail) works as an email client. It connects to the Gmail server and also to other mail servers using the **POP3** or **IMAP** protocol. It shows the mailbox in many panes (parts) so that we can read mails easily. It also has a **junk mail filter** that stops unwanted mail.

Things needed to send an email (2.3.1)

1. An email address
2. A password
3. The email address of the person to whom we want to send the email
4. An Internet connection

Important terms

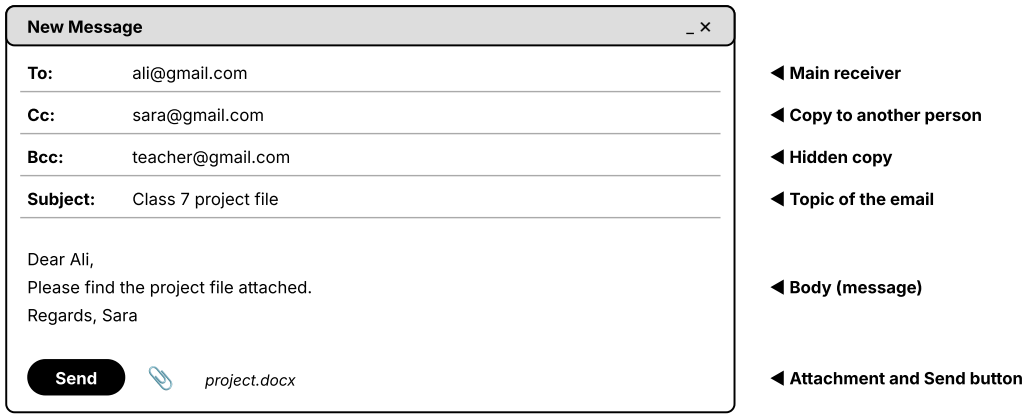
1. **Email Address:** It is the unique address of an email account. Emails are sent to and received at this address. *Example:* testclass7@gmail.com. It has a user name, the @ sign and the domain name.
2. **Password:** It is a secret word, made of letters, numbers and symbols. It protects our account so that no one else can open it.
3. **Recipient:** The person who receives the email is called the recipient. His or her address is written in the To, Cc or Bcc box.
4. **Attachment:** A computer file that is sent along with an email message is called an attachment.
5. **Contacts:** The saved list of names and email addresses of people with whom we communicate.

Write and send an email (2.3.2)

1. Click the **Compose** button in the left panel.
2. Enter the recipient's address.

- **To:** It has the address of the main (primary) recipient.
- **Cc (Carbon Copy):** It sends a copy of the email to another (secondary) address.
- **Bcc (Blind Carbon Copy):** It sends a copy to someone, but other recipients do not know about it.
- If the recipient is in the Address Book, click To, Cc or Bcc to open the Select Names box.
- To send the email to more than one person, put a **semicolon (;)** between the addresses.

1. Write the subject in the **Subject** line.
2. Type the message in the **body** of the email.
3. Click the **Send** button.



Parts of a new email message

Send and receive emails with attachments (2.3.3)

An attachment is usually shown in the reading pane, under the subject or in the message body. Double-click the attachment icon to open it. Files that can be attached:

Types of files we can attach

File type	Full name
bmp	Bitmap Image File
gif	Graphical Interchange Format File
jpg / jpeg	JPEG Image
pdf	Portable Document Format File
doc / docx	Microsoft Word Document
xls / xlsx	Microsoft Excel Spreadsheet
csv	Comma-Separated Values
zip	Data Compression File

Good habits for writing an email

The tone and language of an email should be polite, clear and correct.

1. Use a clear and professional subject line.
2. Proofread (check) every email before you send it.
3. Write the email first, then enter the recipient's address.
4. Double-check that you have the correct recipient.
5. Make sure you Cc all the people who need it.
6. You do not always have to "Reply all".
7. Reply to your emails.
8. Add a signature block at the end.
9. Use the right level of formality (formal or informal).
10. Keep emails short and to the point.

Types of emails that we should remove from the list

Type	Meaning
Bounced	An email that cannot be delivered, or has a delivery problem
Malformed	An email address in the wrong format that does not meet standards
Fake	A fake address made by a fake email generator
Inactive	An address that receives newsletters but never reads them
Service	An address like support@company.com or team@company.com
Unsubscribed	An address whose owner has refused to receive your messages
Duplicate	An email address written in the list more than one time
Spam Trap	An address specially made to catch spammers

Organize emails using folders and labels (2.3.4)

Folders help us to arrange email messages, calendars, contacts and tasks. In Gmail they are called **labels**. To add a folder: in the left pane, **right-click** where you want the folder and click **New Folder**. A new folder is made and we can rename it.

Log in to an email inbox (2.3.5)

1. Go to the **mail.google.com** page.
2. Click the Gmail **Sign in** option.
3. Enter the email address and select **Next**.
4. Enter the password and click the **Next** button.

Long Question: Write the proper protocol of signing out the email account when not using it. ***[Exercise (Brief Q4)]***

1. **Appropriate Use:** After using email, we should always **sign out (log out)** of our account, especially when we use a computer that is not our own, such as a school lab or a cafe computer.
2. **Reason:** If we do not sign out, another person can open our account, read our emails or send wrong emails in our name.
3. Steps to sign out of Gmail:

- Click on the user name (profile picture) at the top of the Gmail page. Some menu options are shown.
- From these options, click the **Sign out** button.

1. **Other good habits:** Never share your password. Do not let the browser save your password on a shared computer.

TOPIC 7: EMAIL AUTHENTICATION, USES, PLATFORMS AND PROTOCOLS: Authentication means checking that a user or an email is real and is coming from the right person.

Process of authentication (2.3.11)

Email authentication is a technical way to check that an email is **not forged (fake)**. It proves that the email really comes from the person it claims to be from. It is mostly used to block harmful and fraud emails such as **spam**.

Long Question: State the different ways in which we can use email as a mean of authentication for another website. ***[Exercise (Brief Q2)]***

Many websites use our email address to check who we are. When we make an account or log in on another website, the website sends a link or a code to our email. This proves that the email is really ours.

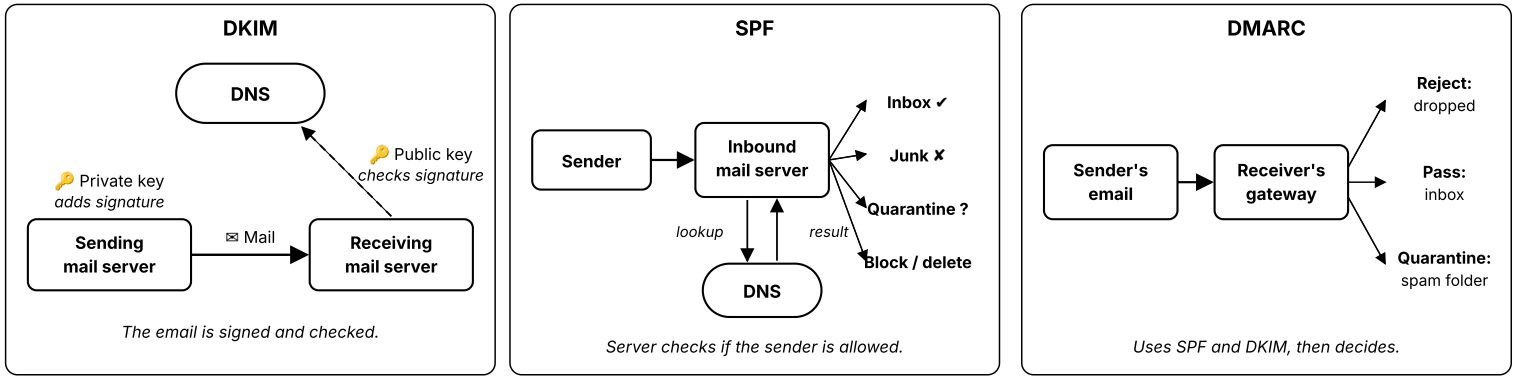
Three common authentication types are:

1. **Password-based Authentication:** The user enters an email address and password. Passwords are the most common method.
2. **Certificate-based Authentication:** A digital certificate is used to identify a user or a device.
3. **Biometric Authentication:** Body features like a fingerprint or face are used.

(The book summary also names **Multi-factor authentication**, where more than one check is used, such as a password plus a code, and **Token-based authentication**.)

Three major methods used in email authentication: All of them are based on **DNS TXT records**.

1. **DKIM (DomainKeys Identified Mail):** The sending mail server adds a secret signature to the email with a **private key**. The receiving server checks it with a **public key** kept in the DNS.
2. **SPF (Sender Policy Framework):** The receiving mail server checks from the DNS whether the sender is allowed to send emails. Then the email goes to the inbox or junk folder, or it is quarantined or blocked.
3. DMARC (Domain-based Message Authentication, Reporting and Conformance): It uses SPF and DKIM. If the email passes, it goes to the inbox. If it fails, it is rejected (dropped) or sent to the spam folder (quarantine). Reports are sent back to the sender.



Three methods of email authentication

<u>Long Question: Discuss the importance and uses of email.</u> ***[Exercise (Brief Q3)]***

Importance: Today people want things to be fast and they want to save time. Email is quick, cheap and easy, so it is used and recommended as a mode of communication all around the world.

Uses of email (2.3.8):

1. We can communicate with people all over the world for free.
2. We can contact more than one person at a time by sending group mails.
3. We can keep a record of documents and conversations.
4. We can work together (in collaboration) with others.
5. We can send files as attachments.
6. We can keep one conversation together for many people.
7. We can create calendars and appointments.

<u>Long Question: Identify and explain the common platforms for electronic mail.</u> ***[Exercise (Brief Q8)]***

An email platform is a website or service that gives us an email account. Some common platforms are:

1. **Gmail:** The free email service of Google. It is the most used platform.
2. **Microsoft Outlook:** The email service of Microsoft (Outlook.com).
3. **Yahoo Mail:** The free email service of Yahoo.
4. **AOL Mail:** An old and well-known free email service.
5. **Zoho Mail:** An email service mostly used by businesses.
6. **ProtonMail:** An email service known for its strong security and privacy.

With these platforms we can send, receive and organize emails from any computer or mobile phone that has the Internet.

Long Question: Create an email address using common freely available platforms like Microsoft or Google. ***[Exercise (Detail Q2)]***

With a free Gmail account, we can use our email, calendar, tasks and contacts from anywhere with an Internet connection. Steps to create a Gmail account (2.3.10):

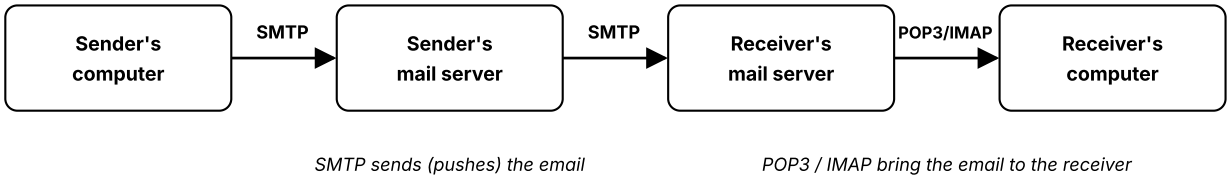
1. Open a web browser, go to the gmail.com sign-up screen and select **Create account**.
2. Select the **"For myself"** option. A new window opens.
3. Enter your first name, last name and user name.
4. Enter a **password**, confirm it and press **Next**. (Use 8 or more characters with letters, numbers and symbols.)
5. Enter a phone number and recovery email address. These fields are optional.
6. Enter your birthday.
7. Select your gender from the drop-down list and click **Next**.
8. Read the privacy policy and terms, and click **"I agree"**.
9. Gmail sets up your account and shows a welcome screen. Now you can use the account on computers and mobile phones.

Long Question: Explain the standard protocols like SMTP and POP3. ***[Exercise (Brief Q10)]***

Email protocol: It is a set of rules that makes sure emails can be exchanged between different servers and email clients in a standard way. This makes email universal, so it works for all users. The common protocols are:

1. **POP (Post Office Protocol):** Email clients use POP to **download** emails from the server. It is mainly a **one-way** protocol. It does not sync (copy) changes back to the server. **POP3** is its version 3.
2. **IMAP (Internet Message Access Protocol):** It is a standard protocol for **incoming** emails. It keeps emails **on the mail server** and lets the user see and manage them as if they were saved on the device. Changes stay the same on all devices.
3. **SMTP (Simple Mail Transfer Protocol):** It is the main protocol for **sending** emails. It transfers emails between email clients and email servers.

In short: SMTP sends the email. POP3 or IMAP brings the email to the receiver.



How SMTP and POP3/IMAP work

SHORT QUESTIONS — UNIT 2

1. Describe the purpose of a word processor. ***[Exercise]***

A word processor is used to type, edit, format, save and print text. It helps us to make letters, reports, tests and homework.

2. Write the four functions of a word processor. ***[Book concept]***

The four functions are composing, editing, saving and printing.

3. Give any four examples of word processors. ***[Book concept]***

MS Word, Google Docs, WordPerfect and LibreOffice Writer.

4. Write a note on benefits of Microsoft PowerPoint. ***[Exercise]***

PowerPoint keeps the talk focused on main points, gives attractive visuals, adds resources from the Internet and explains hard ideas simply. It allows a digital signature, can be changed into video and is easy to share by email.

5. State the ways in which we can use email for authentication on another website. ***[Exercise]***

Websites send a link or code to our email to prove who we are. Common types are password-based, certificate-based and biometric authentication. Email is checked with DKIM, SPF and DMARC.

6. Discuss the importance and uses of email. ***[Exercise]***

Email is fast, cheap and easy. We use it to talk to people all over the world for free, send group mails, send attachments, work together and make calendars and appointments.

7. Write the proper protocol of signing out the email account. ***[Exercise]***

Click on the user name at the top of the Gmail page and then click the Sign out button. We should always sign out on a computer that is not our own.

8. Elaborate on the use of thesaurus and synonyms in MS Word. ***[Exercise]***

The thesaurus finds synonyms (same meaning) and antonyms (opposite meaning) of a word. Select a word, right-click, go to Synonyms and click the best word.

9. Differentiate between Save and Save As. ***[Exercise]***

Save updates the same file with new work (Ctrl + S). Save As saves the file for the first time or with a new name, place or format (F12).

10. Identify the common platforms for electronic mail. ***[Exercise]***

Gmail, Microsoft Outlook, Yahoo Mail, AOL, Zoho and ProtonMail.

11. Define recipient. ***[Exercise]***

The person who receives an email is called the recipient.

12. Define attachment. ***[Exercise]***

A computer file that is sent along with an email message is called an attachment.

13. Define password. ***[Exercise]***

A password is a secret word of letters, numbers and symbols that protects our account.

14. Define email address. ***[Exercise]***

It is the unique address of an email account where emails are sent and received, such as testclass7@gmail.com.

15. Explain the standard protocols SMTP and POP3. ***[Exercise]***

SMTP (Simple Mail Transfer Protocol) sends emails between clients and servers. POP3 (Post Office Protocol) downloads emails from the server to the user's device.

16. What is the difference between POP3 and IMAP? ***[Book concept]***

POP3 downloads emails and does not sync back to the server. IMAP keeps emails on the server, so they look the same on all devices.

17. What is the difference between Cc and Bcc? ***[Book concept]***

Cc sends a copy that all recipients can see. Bcc sends a hidden copy that other recipients do not know about.

18. What is a multimedia presentation? ***[Book concept]***

It is a set of media objects such as text, pictures, video and sound shown on slides. The easiest way to make it is MS PowerPoint.

19. What are speaker notes? ***[Book concept]***

Speaker notes are notes added to a slide that help the presenter remember the main points. The audience does not see them.

20. What is the clipboard? ***[Book concept]***

The clipboard is a temporary storage area where cut or copied text and images are kept.

21. Differentiate between numbered and bulleted lists. ***[Book concept]***

A numbered list has numbers or letters and is used when order matters. A bulleted list has bullets and is used when order does not matter.

22. What are header and footer? ***[Book concept]***

A header appears in the top margin and a footer appears in the bottom margin of every page. They show the title, date or page numbers.

23. What are page margins? ***[Book concept]***

Margins are the blank spaces on the top, bottom, left and right sides of a page.

24. What is email authentication? ***[Book concept]***

It is a technical way to check that an email is not fake and comes from the real sender. It blocks spam.

25. What things do we need to send an email? ***[Book concept]***

An email address, a password, the receiver's email address and an Internet connection.

26. Write the full forms of DKIM, SPF and DMARC. ***[Book concept]***

DKIM: DomainKeys Identified Mail. SPF: Sender Policy Framework. DMARC: Domain-based Message Authentication, Reporting and Conformance.

MULTIPLE CHOICE QUESTIONS — UNIT 2

1. To open a word document, we go to the __ tab. ***[Exercise MCQ 1]***

a) Open ✔ **b) File** c) Home d) Insert

2. __ is a word processor that allows us to enter, format, save and print text. ***[Exercise MCQ 2]***

a) MS Excel b) MS PowerPoint ✔ **c) MS Word** d) MS Paint

3. We can format the text from the __ tab. ***[Exercise MCQ 3]***

a) File b) Format c) Insert ✔ **d) Home**

4. __ makes the text appear thicker and darker. ***[Exercise MCQ 4]***

a) Italics b) Format c) Font face ✔ **d) Bold**

5. There are __ types of alignments in text formatting in Word. ***[Exercise MCQ 5]***

a) 3 ✔ **b) 4** c) 5 d) 6

6. The __ tab lets you control the look and feel of your document in MS Word. ***[Exercise MCQ 6]***

✔ **a) Layout** b) Margin c) Caption d) Format

7. We can insert the image in the Word document from the __ tab. ***[Exercise MCQ 7]***

✔ **a) Insert** b) File c) Home d) View

8. We can resize the image by clicking and dragging on its __. ***[Exercise MCQ 8]***

a) Outline | b) Center | ✔ **c) Resize handles** | d) Arrows

9. The keyboard shortcut to copy is __. ***[Exercise MCQ 9]***

a) Ctrl + X | ✔ **b) Ctrl + C** | c) Ctrl + P | d) Ctrl + S

10. We can insert the table in the Word document from the __ tab. ***[Exercise MCQ 10]***

a) File | b) Home | ✔ **c) Insert** | d) View

11. __ appears at the top margin of the Word document. ***[Exercise MCQ 11]***

a) Footer | b) Title | c) Address bar | ✔ **d) Header**

12. To print the Word document, we go to the __ tab. ***[Exercise MCQ 12]***

a) Print | ✔ **b) File** | c) Format | d) Insert

13. The easiest way to create a multimedia presentation is to create it on __. ***[Exercise MCQ 13]***

a) Microsoft Word | b) Microsoft Excel | ✔ **c) Microsoft PowerPoint** | d) Microsoft Paint

14. The keyboard shortcut to create new email is __. ***[Exercise MCQ 14]***

a) Ctrl + S | b) Ctrl + N | ✔ **c) Ctrl + M** | d) Ctrl + X

15. It is always safe to __ your account, if not using it. ***[Exercise MCQ 15]***

a) Protect | ✔ **b) Sign out** | c) Sign in | d) Delete

16. How many primary functions does a word processor have?

a) two | b) three | ✔ **c) four** | d) five

17. The keyboard shortcut to cut is:

a) Ctrl + C | ✔ **b) Ctrl + X** | c) Ctrl + V | d) Ctrl + Z

18. The keyboard shortcut to paste is:

a) Ctrl + P | b) Ctrl + C | ✔ **c) Ctrl + V** | d) Ctrl + X

19. Cut or copied text is kept in the:

a) Recycle Bin | ✔ **b) Clipboard** | c) Ribbon | d) Title Bar

20. The shortcut key for spelling and grammar check is:

a) F5 | ✔ **b) F7** | c) F12 | d) F2

21. The shortcut key for Save is:

✔ **a) Ctrl + S** | b) Ctrl + A | c) F7 | d) F5

22. The shortcut key for Save As is:

a) F5 | b) F7 | ✔ **c) F12** | d) Ctrl + V

23. Which command lets us change the name, location and format of a file?

a) Save | ✔ **b) Save As** | c) Open | d) Print

24. Margins are the blank spaces on the __ of a page.

a) top only | b) left and right only | ✔ **c) top, bottom, left and right** | d) middle

25. Footer appears at the __ margin of the page.

a) top | ✔ **b) bottom** | c) left | d) right

26. The tool used to find synonyms and antonyms is:

a) Spell check | ✔ **b) Thesaurus** | c) Clipboard | d) Ribbon

27. Words with the same meaning are called:

a) antonyms | ✔ **b) synonyms** | c) headers | d) bullets

28. To stop a bulleted list, press the Enter key:

a) once | ✔ **b) twice** | c) three times | d) four times

29. The Pictures button is in the __ group.

a) Font | b) Paragraph | ✔ **c) Illustrations** | d) Clipboard

30. Which orientation makes the page wide?

a) Portrait | ✔ **b) Landscape** | c) Margin | d) Column

31. To start a slide show from the first slide, press:

a) F7 | b) F12 | ✔ **c) F5** | d) F2

32. Speaker notes are added from the __ tab.

a) Insert | ✔ **b) View** | c) Home | d) Design

33. The slide show option that works best on a CD is:

✔ **a) Browsed by an individual (window)** | b) Browsed at a kiosk (full screen) | c) Rehearse Timings | d) Slide Sorter

34. Time for each slide is recorded using:

✔ **a) Rehearse Timings** | b) Spell check | c) Thesaurus | d) Save As

35. Cc stands for:

a) Copy Click | ✔ **b) Carbon Copy** | c) Clear Copy | d) Central Copy

36. Bcc stands for:

a) Blind Clear Copy | ✔ **b) Blind Carbon Copy** | c) Black Carbon Copy | d) Basic Carbon Copy

37. To send an email to more than one person, we put a __ between addresses.

a) comma | b) full stop | ✔ **c) semicolon** | d) colon

38. A file sent along with an email is called a/an:

a) subject | ✔ **b) attachment** | c) signature | d) label

39. In Gmail, folders are also called:

a) tabs | ✔ **b) labels** | c) slides | d) pages

40. The protocol used to send emails is:

a) POP3 | b) IMAP | ✔ **c) SMTP** | d) DNS

41. SMTP stands for:

✔ **a) Simple Mail Transfer Protocol** | b) Simple Message Text Protocol | c) Secure Mail Transfer Protocol | d) Simple Mail Text Program

42. POP stands for:

a) Public Office Protocol | ✔ **b) Post Office Protocol** | c) Post Online Protocol | d) Private Office Program

43. The protocol that keeps emails on the mail server is:

a) POP3 | b) SMTP | ✔ **c) IMAP** | d) SPF

44. Which is NOT an email platform?

a) Gmail | b) Yahoo Mail | c) ProtonMail | ✔ **d) PowerPoint**

45. DKIM, SPF and DMARC are all based on __ records.

a) Excel | ✔ **b) DNS TXT** | c) PDF | d) ZIP

46. The most common method of authentication is:

a) biometric | b) certificate | ✔ **c) password** | d) token

47. An email address made specially to catch spammers is called a:

a) bounced email | b) duplicate email | ✔ **c) spam trap** | d) service email

Mistakes Found in the Book

1. Page 110 answer key, Unit 2 MCQ 5: The key gives (a) 3. But the book itself (page 20) shows **four** alignments: left, center, right and justify. So the correct answer is **(b) 4**.
2. Page 40, MCQ 14: The book does not teach this shortcut anywhere. The answer key gives (c) Ctrl + M, so these notes tick (c) for the exam. In real Outlook, a new email is made with **Ctrl + N** (or Ctrl + Shift + M).
3. Page 34 says there are **3** common authentication types, but the summary on page 39 says there are **5** (it adds multi-factor and token-based). These notes give the 3 types and mention the other 2.
4. Page 33: The heading says "Log in an email inbox" and the text talks about **Outlook**, but all the steps are for **Gmail** (mail.google.com).

Computer Science — Class 7

Single National Curriculum (SNC) — Punjab Curriculum and Textbook Board

Unit 3 — Computational Thinking

TOPIC 1: COMPUTATIONAL THINKING: Computational Thinking (CT) is a set of problem-solving skills and methods that we use to solve a problem.

What is computational thinking? (3.1)

Computational thinking helps us to take a difficult (complex) problem and understand what the problem is. Then we make possible solutions for it. We present these solutions in a way that a computer, a human, or both can understand.

In the book, a Venn diagram shows that computational thinking and computer science are both parts of **computing**, and they overlap with each other.

Cornerstones (techniques) of computational thinking

There are four key techniques of computational thinking:

1. **Decomposition:** Breaking a task or problem into small steps or parts.
2. **Pattern Recognition:** Finding things that are similar between problems.
3. **Generalization and Abstraction:** Finding the main rules (principles) that cause these patterns.
4. **Algorithm Design:** Writing step-by-step instructions to solve the problem, and to solve similar problems again.

Extra Bit: Interdisciplinary studies mean joining two or more subjects into one activity.

Long Question: Define pattern recognition in your words. ***[Exercise (Detail Q1)]***

1. **Meaning:** Pattern recognition means looking for patterns (similar things) in a problem. We check whether a problem or a solution that we have seen before can be used here.
2. **What we look for:** It may also mean recognizing similar shapes, sounds or pictures.
3. **Use:** It is used to find similarities between the small problems that we get after decomposition.
4. **Example:** When we make a computer game, we find similar objects, patterns and actions in it. Then we can use the same (or a slightly changed) method for all of them. This makes our solution faster and better (more efficient).

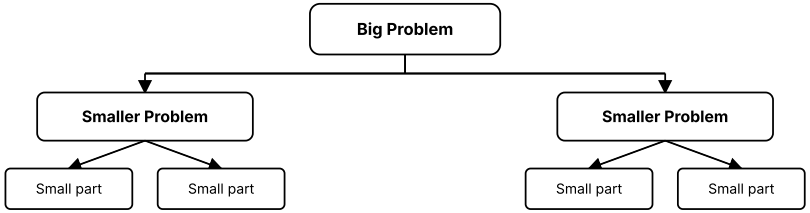
Do you know? Patterns are the laws of nature and life. They are found in all subjects of life.

Decomposition

Decomposition means breaking a big and difficult problem into smaller parts that are easy to manage. Small problems are easy to solve. Decomposition is an important skill of life.

Example: If we want to host a holiday dinner, we can break the task like this:

- Select the menu.
- Get help from others in the kitchen.
- Tell people what to bring.
- Decide how each dish will be cooked.
- Set the time for the event.



Decomposition: a big problem is broken into smaller problems

Generalization and abstraction

1. **Abstraction:** It helps us to pick the details that are **important** for solving the problem and to **ignore** the details that are not important.
2. **Generalization:** It helps us to make a **general idea** of the problem and how to solve it. After finding patterns, we write them in the simplest form so that we can use them again whenever needed.

Example: A car and a bicycle are both **wheeled vehicles**, and a boat and a wheeled vehicle are both **vehicles**. We ignore small details like colour and size and keep only the main idea.

TOPIC 2: ALGORITHMS: An algorithm is a set of instructions for solving a problem or doing a task.

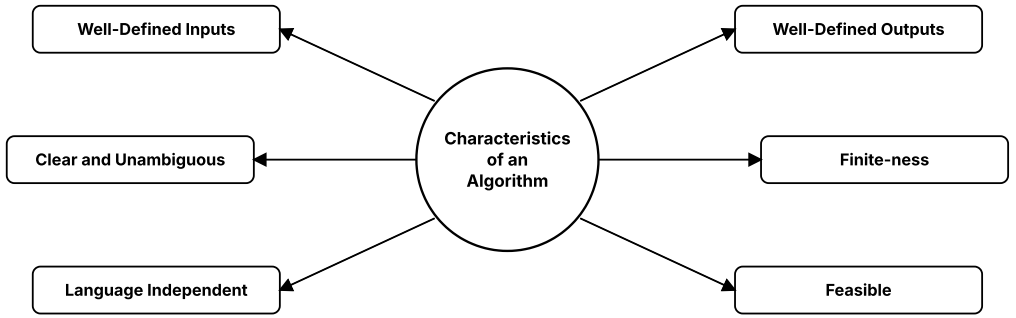
Algorithm design

An algorithm is a sequence (series) of steps that a human or a computer can follow to complete a task. It is a step-by-step method that gives the answer to a question, or the solution of a problem, in a **finite (limited) number of steps**.

Characteristics of an algorithm

1. **Clear and Unambiguous:** Each step must be clear and must have only one meaning.
2. **Well-Defined Inputs:** The inputs of the algorithm must be clearly defined.

3. **Well-Defined Outputs:** The algorithm must clearly tell what the output will be.
4. **Finite-ness:** The algorithm must end (stop) after a limited time.
5. **Feasible:** The algorithm must be simple and practical, so it can be done with the available resources.
6. **Language Independent:** The algorithm must not depend on any programming language.



Characteristics of an algorithm

Types of algorithms

1. **Brute Force Algorithm:** It is the simplest method. It tries **all possible solutions** one by one until the right solution is found.
2. **Recursive Algorithm:** The problem is broken into several sub-parts, and the **same procedure is called again and again**.
3. **Searching Algorithm:** It is used to **search** an item, or a group of items, from given data.
4. **Sorting Algorithm:** It is used to **arrange** a group of data in a particular order (manner), such as smallest to largest.
5. **Divide and Conquer Algorithm:** It breaks a problem into sub-problems, solves each sub-problem and then **merges (joins)** the solutions to get the final solution.
6. **Randomized Algorithm:** It uses a **random number** to give an immediate benefit. The random number helps to decide the expected result.

Structures of an algorithm

1. **Sequence Structure:** The steps are carried out in the **same order** as they are written.
2. **Selection (Conditional) Structure:** A step, or a set of steps, is carried out **only if a condition is true**. Otherwise, the **else** part is carried out. *Example:* We use an "if then else" condition to find whether a number is even or odd.
3. **Repetition (Loop) Structure:** A **loop** is a set of instructions that is carried out again and again while a certain condition is true. When the condition becomes false, the loop stops and the steps after the loop are carried out.

Types of loops

1. **Finite Loop:** It is the most common type of loop. It has a clear end. It runs its steps a **fixed number of times** and stops when the condition becomes false.
2. **Infinite Loop:** It has **no clear end**. It runs forever because its condition always remains true. *Example:* The **"forever"** block in Scratch makes a sprite repeat its action forever.

Computational thinking in real life

Computational thinking is seen in the way people make decisions, do basic maths and solve problems. *Examples:*

- Using an algorithm to find the best route from school to home, keeping in mind traffic, construction work or road blocks.
- Following a recipe to bake a cake is an example of an algorithm.
- Planning a budget uses pattern recognition and decomposition when we check our spending in each category.

Essentials (pre-requisites) of writing an algorithm

Before writing an algorithm, we must know these five things:

1. **The Problem:** A clear definition of the problem to be solved.
2. **The Limitations (Constraints):** The limits that we must keep in mind while solving the problem.
3. **The Input:** The data that we take to solve the problem.
4. **The Output:** The result that we expect when the problem is solved.
5. **The Process (Solution):** The method of solving the problem within the given limits.

Then we write the algorithm with the help of these five things.

Long Question: Write an algorithm to find sum of three numbers. ***[Exercise (Detail Q2)]***

Step 1: Pre-requisites

1. **Problem:** Add three numbers and print their sum.
2. **Constraints:** The numbers must have only digits and no other characters.
3. **Input:** The three numbers to be added.
4. **Output:** The sum of the three numbers (a single integer value).
5. **Solution:** Add the three numbers with the help of the '+' operator.

Step 2: Algorithm

Algorithm: Sum of three numbers
1. START
2. Declare 3 integer variables num1, num2 and num3.
3. Take the three numbers as input in num1, num2 and num3.
4. Declare an integer variable sum.
5. sum = num1 + num2 + num3
6. Print the value of sum.
7. END

Long Question: Write the algorithm to find whether given number is even or odd. ***[Exercise (Detail Q3)]***

Step 1: Pre-requisites

1. **Problem:** Print whether a number is even or odd.
2. **Constraints:** The number must have only digits and no other characters.
3. **Input:** A number to be checked.
4. **Output:** Print "even" or "odd".
5. **Solution:** Divide the number by 2 and take the remainder. If the remainder is zero, the number is even. Otherwise it is odd.

Step 2: Algorithm

Algorithm: Even or odd
1. START
2. Declare an integer variable num1.
3. Take the number as input in num1.
4. Declare an integer variable remainder.
5. remainder = num1 % 2 (% gives the remainder)
6. IF remainder = 0, print "Number is Even".
Otherwise print "Number is Odd".
7. END

Extra Bit: "Mod" (%) is used to take the remainder (modulus) of an integer. *Example:* 7 % 2 = 1 and 8 % 2 = 0.

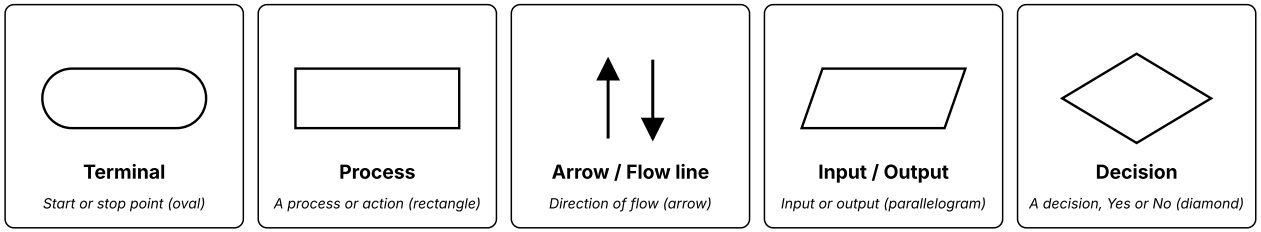
TOPIC 3: FLOWCHARTS: A flowchart is a picture (symbolic) form of an algorithm.

What is a flowchart?

Each step in a flowchart is shown by a **symbol** with a short description of the step. **Arrows** join the steps and show the direction (flow) of the flowchart. Flowcharts are used to study, design, document and manage a process or a program.

Commonly used symbols in flowcharts

Symbol	Name	Use
Oval	Terminal / Terminator	Shows the start or stop point of a flowchart
Rectangle	Action / Process	Shows a process (work) in the flowchart
Arrow	Arrow / Flow line	Shows the direction of flow from one step to another
Parallelogram	Input / Output	Shows input or output
Diamond	Decision	Shows a decision (Yes or No)



Flowchart symbols

Long Question: Write down rules of writing a flowchart. ***[Exercise (Detail Q4)]***

We should follow these rules while drawing a flowchart:

1. Use the standard (conventional) flowchart symbols.
2. Every flowchart must have a **Start** and an **End (Stop)** point.
3. Each symbol should have only **one exit point**, except the decision symbol (it has two: Yes and No).
4. Flow lines should **not cross** each other.
5. The flow lines coming out of a decision symbol should be **labelled** properly (Yes / No).

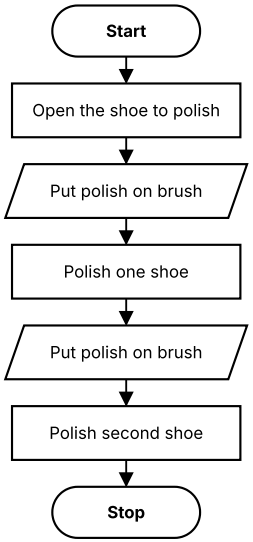
Advantages and disadvantages of flowcharts

Advantages	Disadvantages
Easy to make	Difficult to show big (complex) programs
Easy to understand	Changing (modifying) it is difficult
Mistakes can be found easily	It takes a lot of time
Debugging becomes possible (removing errors is called debugging)	Difficult to understand for people who do not know the symbols
The logic can be understood easily	

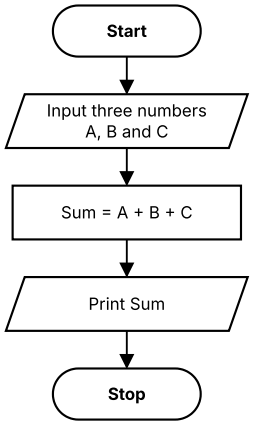
Long Question: Draw a flowchart to find sum of three numbers. ***[Exercise (Detail Q6)]***

The flowchart takes three numbers A, B and C as input, adds them and prints the sum (Example 2 below). Example 1 shows a simple flowchart of a daily life task: polishing shoes.

Example 1: Polish your shoes



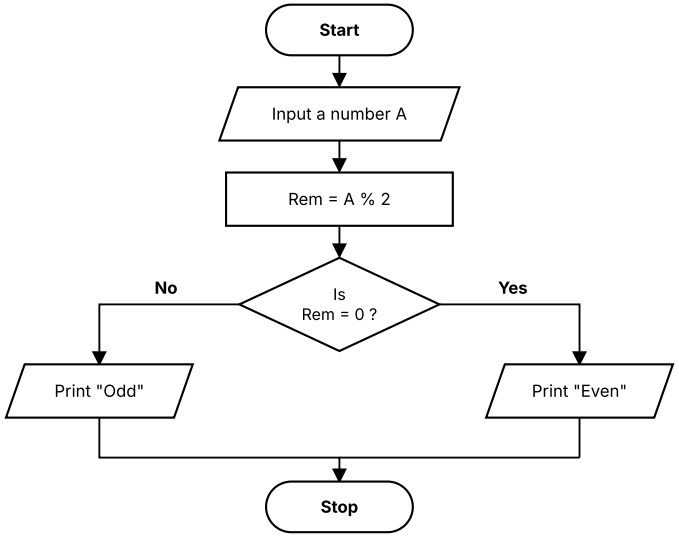
Example 2: Sum of three numbers



Example 1: Polish your shoes; Example 2: Sum of three numbers

Long Question: Draw a flowchart to find whether given number is even or odd. ***[Exercise (Detail Q7)]***

The flowchart takes a number A, finds the remainder by dividing it by 2 (Rem = A % 2) and checks the remainder with a decision symbol. If Rem = 0, it prints "Even". Otherwise it prints "Odd".

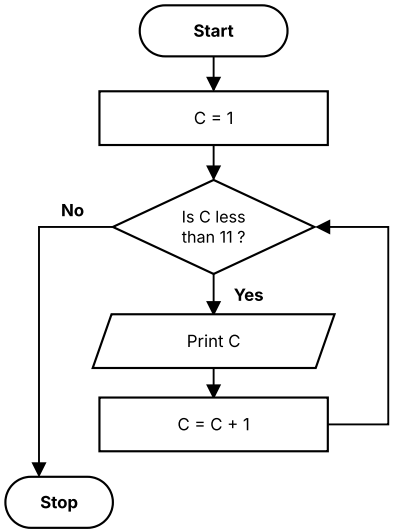


Example 3: Flowchart to find whether a number is even or odd

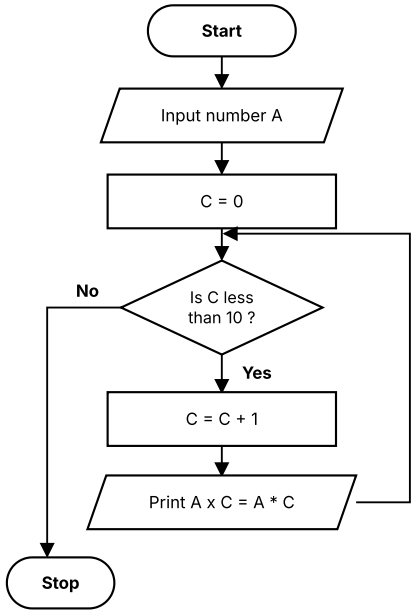
Flowcharts with loops

Example 4 prints the numbers 1 to 10. It starts with C = 1 and repeats while C is less than 11. Example 5 prints the table of a number A up to 10. Both use a **finite loop**.

Example 4: Print first 10 numbers



Example 5: Table of a number up to 10



Example 4: Print first 10 numbers; Example 5: Table of a given number up to 10

TOPIC 4: EFFICIENCY OF A SOLUTION: One problem can have more than one solution. One solution can be more efficient (better) than the others.

Long Question: How can we measure the efficiency of a solution? ***[Exercise (Detail Q8)]***

Efficiency of a solution is measured by two things:

1. **Number of Steps:** How many steps the solution takes to solve the problem.
2. **Memory:** How much memory the solution uses.

Efficiency is mostly measured by the **number of steps**. The solution that solves the problem in **fewer steps** is more efficient.

Example: Find the largest number from three numbers.

Solution 1 (7 steps)

1. Start.
2. Take three different numbers in A, B and C.
3. Set largest = A.
4. If B is greater than largest, set largest = B.
5. If C is greater than largest, set largest = C.
6. Print largest.
7. Stop.

Solution 2 (8 steps)

1. Start.
2. Input three different numbers A, B and C.
3. If A is greater than B, go to step 6.
4. If B is greater than C, print "B is largest" and go to step 8.
5. Print "C is largest" and go to step 8.
6. If A is greater than C, print "A is largest" and go to step 8.
7. Print "C is largest".
8. End.

Analysis: Both solutions solve the same problem. Solution 1 takes **7 steps** and uses four variables (A, B, C and largest). Solution 2 takes **8 steps** and uses three variables (A, B and C). Solution 1 takes fewer steps, so it is **more efficient**, although Solution 2 uses less memory.

Long Question: Why don't we consider memory in measuring efficiency of an algorithm? ***[Exercise (Detail Q5)]***

1. **Two parameters:** The efficiency of a solution can be measured by the number of steps and by the memory it uses.
2. **Memory is not a problem:** Modern computers have a lot of memory, and memory is cheap. So memory is **not an issue** for computer systems now.
3. **Steps matter more:** The number of steps decides how fast the problem is solved. So efficiency is always measured in terms of the **number of steps**.
4. **Example:** In the largest-number example, Solution 1 uses one extra variable (more memory) but takes fewer steps. Still, Solution 1 is called more efficient.

SHORT QUESTIONS — UNIT 3

1. Define computational thinking in your words.

[Exercise]

Computational thinking is a set of problem-solving skills. It helps us to understand a difficult problem and make solutions that a computer or a human can understand.

2. Enlist techniques of computational thinking.

[Exercise]

The techniques are decomposition, pattern recognition, generalization and abstraction, and algorithm design.

3. What do you mean by decomposition?

[Exercise]

Decomposition means breaking a big and difficult problem into smaller parts that are easy to solve. *Example:* dividing the work of a holiday dinner into small tasks.

4. Elaborate generalization and abstraction. ***[Exercise]***

Abstraction means keeping the important details and ignoring the unimportant ones. Generalization means making a general idea of the problem and its solution so that it can be used again.

5. Define algorithm in your words. ***[Exercise]***

An algorithm is a set of step-by-step instructions to solve a problem or do a task in a limited number of steps.

6. Enlist the characteristics of an algorithm. ***[Exercise]***

Clear and unambiguous, well-defined inputs, well-defined outputs, finite-ness, feasible and language independent.

7. Differentiate between recursive and brute force algorithm. ***[Exercise]***

A brute force algorithm tries all possible solutions until the right one is found. A recursive algorithm breaks the problem into sub-parts and calls the same procedure again and again.

8. Differentiate between searching and sorting algorithm. ***[Exercise]***

A searching algorithm finds an item or a group of items in data. A sorting algorithm arranges data in a particular order.

9. What are loops? Discuss its types. ***[Exercise]***

A loop is a set of instructions that is repeated again and again while a condition is true. Its types are finite loop (stops after a fixed number of times) and infinite loop (never stops).

10. Enlist advantages and disadvantages of flowcharts. ***[Exercise]***

Advantages: easy to make and understand, mistakes are found easily, debugging is possible. Disadvantages: hard for big programs, hard to change, takes time.

11. What is a flowchart? ***[Book concept]***

A flowchart is a picture form of an algorithm. Each step is shown by a symbol and arrows show the flow.

12. What is pattern recognition? ***[Book concept]***

It means finding similar things (patterns) in problems, so that an old solution can be used again.

13. What is a divide and conquer algorithm? ***[Book concept]***

It breaks a problem into sub-problems, solves them and merges the solutions to get the final solution.

14. What is a randomized algorithm? ***[Book concept]***

It is an algorithm that uses a random number to give an immediate benefit.

15. What is sequence structure? ***[Book concept]***

In sequence structure, the steps are carried out in the same order as they are written.

16. What is selection structure? ***[Book concept]***

In selection structure, a step is carried out only if a condition is true. Otherwise the else part is carried out.

17. Differentiate between finite and infinite loop. ***[Book concept]***

A finite loop runs a fixed number of times and then stops. An infinite loop never stops because its condition is always true.

18. What is debugging? ***[Book concept]***

The process of finding and removing errors is called debugging.

19. Write the pre-requisites of writing an algorithm. ***[Book concept]***

The problem, its limitations (constraints), the input, the expected output and the process (solution).

20. Which symbol is used for decision in a flowchart? ***[Book concept]***

The diamond symbol is used for a decision.

21. Give two real life examples of computational thinking. ***[Book concept]***

Following a recipe to bake a cake, and finding the best route from school to home.

22. What does the % (mod) operator do? ***[Book concept]***

It gives the remainder of a division. *Example:* 7 % 2 = 1.

MULTIPLE CHOICE QUESTIONS — UNIT 3

1. Breaking down a problem into sub problems is called: ***[Exercise MCQ 1]***

a) generalization	b) pattern Recognition	✔ **c) deconstruction**	d) Design

2. Discover the principles that cause the patterns of a problem is called: ***[Exercise MCQ 2]***

✔ **a) generalization**	b) pattern Recognition	c) deconstruction	d) Design

3. Set of instructions to solve a problem is called: ***[Exercise MCQ 3]***

a) directions	b) instructions	✔ **c) algorithm**	d) Design

4. The algorithm which goes through all possible solutions until required solution is found is called: ***[Exercise MCQ 4]***

a) Recursive Algorithm	b) Searching algorithm	✔ **c) Brute force algorithm**	d) sorting algorithm

5. The algorithms which help in arranging a group of data in a particular manner are called: ***[Exercise MCQ 5]***

a) Recursive Algorithm	b) Searching algorithm	c) Brute force algorithm	✔ **d) sorting algorithm**

6. The algorithm which breaks a problem into sub-problems, solves a single sub-problem and merges the solutions is called: ***[Exercise MCQ 6]***

a) Recursive Algorithm	b) Brute force algorithm	✔ **c) divide and conquer Algorithm**	d) sorting algorithm

7. The algorithm which uses a random number so that it gives immediate benefit is called: ***[Exercise MCQ 7]***

a) Recursive Algorithm	b) divide and conquer Algorithm	c) Brute force algorithm	✔ **d) random algorithm**

8. The sequence where we repeat specific set of instruction again and again is called: ***[Exercise MCQ 8]***

✔ **a) loop**	b) sequence	c) condition	d) all

9. The loops which have to be terminated are called: ***[Exercise MCQ 9]***

a) Infinite loops	✔ **b) Finite loops**	c) intermediate loops	d) simple loops

10. The loops which are never going to end are called: ***[Exercise MCQ 10]***

✔ **a) Infinite loops**	b) Finite loops	c) intermediate loops	d) simple loops

11. CT stands for:

a) Computer Technology	✔ **b) Computational Thinking**	c) Common Thinking	d) Coding Technique

12. Looking for similar things in problems is called:

a) decomposition ✔ **b) pattern recognition** c) abstraction d) algorithm design

13. Keeping important details and ignoring unimportant ones is called:

a) decomposition b) pattern recognition ✔ **c) abstraction** d) debugging

14. An algorithm must end after a finite time. This characteristic is called:

a) feasible ✔ **b) finite-ness** c) language independent d) unambiguous

15. An algorithm must not depend on any programming language. This is called:

✔ **a) language independent** b) feasible c) finite-ness d) well-defined input

16. The algorithm that calls the same procedure again and again is:

a) brute force b) sorting ✔ **c) recursive** d) searching

17. The algorithm used to find an item from given data is:

a) sorting ✔ **b) searching** c) randomized d) recursive

18. In __ structure, steps are carried out in the same order as written.

a) selection b) loop ✔ **c) sequence** d) decision

19. If-then-else is used in __ structure.

a) sequence ✔ **b) selection** c) loop d) none

20. The "forever" block in Scratch makes a/an:

a) finite loop ✔ **b) infinite loop** c) decision d) input

21. The symbol used for start and stop in a flowchart is:

a) rectangle b) diamond ✔ **c) oval** d) parallelogram

22. The symbol used for input and output in a flowchart is:

a) oval b) rectangle ✔ **c) parallelogram** d) diamond

23. The symbol used for a process in a flowchart is:

✔ **a) rectangle** b) oval c) diamond d) arrow

24. The diamond symbol in a flowchart shows a:

a) process ✔ **b) decision** c) start d) input

25. The process of removing errors is called:

a) decomposition ✔ **b) debugging** c) sorting d) looping

26. Flow lines in a flowchart should not:

a) start ✔ **b) cross each other** c) have arrows d) be labelled

27. The % operator gives the:

a) sum b) product ✔ **c) remainder** d) quotient

28. 8 % 2 is equal to:

a) 4 ✔ **b) 0** c) 1 d) 2

29. Efficiency of a solution is mostly measured by the:

a) memory used ✔ **b) number of steps** c) colour of flowchart d) number of variables

30. Following a recipe to bake a cake is an example of a/an:

✔ **a) algorithm** b) loop only c) debugging d) sorting

Mistakes Found in the Book

1. Page 110 answer key, Unit 3 MCQ 2: The key gives (b) pattern recognition. But the book (page 44 and the summary) says that discovering the principles that cause patterns is **generalization**. So the correct answer is **(a)**.
2. Page 110 answer key, Unit 3 MCQ 6: The key gives (b) brute force algorithm. The book (page 46) calls this the **divide and conquer algorithm**, so the correct answer is **(c)**.
3. Page 110 answer key, Unit 3 MCQ 9: The key gives (a) infinite loops. Loops that end (are terminated) are **finite loops**, so the correct answer is **(b)**.
4. Page 54, MCQ 1: None of the options says "decomposition". The key gives (c) deconstruction, which is the closest word, so these notes tick (c).
5. Page 55, Detail Q1: "pattern reorganization" should be "pattern **recognition**".
6. Page 46: "Lepitaton Structure" should be "**Repetition** Structure". The same page writes "alogorithm" for "algorithm".
7. Page 53, Analysis: It says Solution 1 is more efficient "although it consumed less memory". In fact Solution 1 uses **more** memory (four variables), and Solution 2 uses less memory.
8. Page 46: The text uses the word "member" in "whether a member is even or odd". It should be "**number**".

Computer Science — Class 7

Single National Curriculum (SNC) — Punjab Curriculum and Textbook Board

Unit 4 — Programming

TOPIC 1: COMPUTERS STORE INFORMATION USING BINARY CODES: Computers store everything as numbers. They can show everything by using only two numbers: **0 (zero) and 1 (one)**.

Binary codes (4.1)

1. **Binary Numbers:** Writing numbers as a set of zeros and ones is called using binary numbers.
2. **Switches:** Inside a computer, numbers are shown by electronic switches. An **open switch means 0** and a **closed switch means 1**. With enough switches, almost anything can be shown as numbers.
3. **Bits:** These switches are also called **bits**. Bits are stored in the computer's memory. **Bit** is the short form of **Binary Digit**.
4. **Byte: Eight bits** together make one **byte** of data.
5. **Memory:** A megabyte is about millions of bytes and a gigabyte is about billions of bytes. The amount of memory in a computer decides how many numbers it can store.

Binary explained

The binary number system was improved (refined) in the **17th century** by **Gottfried Leibniz** (1646-1716). He invented the binary system around **1679** and published it in **1701**. It became the base of almost all modern computers. Leibniz also invented calculus.

- A bit is the **smallest unit of data**. It has only one value: either **1 or 0**.
- Computers show numbers in binary code (1s and 0s) inside the **CPU and RAM**. These are electrical signals that are either **on (1)** or **off (0)**.

Long Question: Describe how computers can save information using binary codes. ***[Exercise (Brief Q1)]***

1. **Only two numbers:** A computer stores all information (text, pictures, sound and numbers) as numbers. It uses only two digits, **0 and 1**. This is called **binary code**.
2. **Switches:** Inside the computer there are millions of tiny electronic switches. An **open (off)** switch means **0** and a **closed (on)** switch means **1**.
3. **Bits and Bytes:** Each switch is called a **bit** (binary digit). Eight bits make one **byte**. Bits are stored in the memory (RAM) and used by the CPU.
4. **Electrical signals:** In the CPU and RAM, these 1s and 0s are electrical signals that are either on or off.
5. **Codes for data:** Text is changed into binary with **ASCII** or **Unicode**. Pictures are stored as **pixels** and colours as **RGB** numbers. Sound is changed into numbers with an **ADC**.

Conclusion: The more memory a computer has, the more bits it can store, so it can save more information.

Importance of binary code

The binary number system is the base of **all computing systems and operations**. It helps devices to store, access and change all types of information going to and from the CPU or memory. Because of it, we can:

1. view websites;
2. create and update documents;
3. play games;
4. watch streaming videos and other pictures (graphics);
5. use software; and
6. do calculations and data analysis.

The system of 1s and 0s is a simple and neat way for computers to work. It also controls logic circuits and detects whether an electrical signal is **true (1) or false (0)**.

TOPIC 2: BINARY AND DECIMAL NUMBER SYSTEMS: A number system is a way of writing numbers. Computers use the binary system and people use the decimal system.

Decimal number system (4.2)

1. **Base:** It has **base 10** and uses **10 symbols**: 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.
2. **Other name:** It is also called the **Hindu-Arabic** number system.
3. **Place value:** Each place is **ten times** more than the place on its right. In 36, the 3 is in the tens place, so it means 30.
4. **Decimal point:** It uses a decimal point to show fractions.
5. **Writing:** Decimal numbers are written as 45_{10} or 118_{10}. If the base of a number is **not written**, it is taken as a **decimal number**. It is the most commonly used number system.

1	9	8	0
1000th place	100th place	10th place	1st place

Binary 1010 = Decimal 10

1	0	1	0

$1\times2^3 + 0\times2^2 + 1\times2^1 + 0\times2^0$

= 8 + 0 + 2 + 0 = 10

Place values in 1,980 (decimal) and the value of binary 1010

Binary number system

1. **Base:** It has **base 2** and uses only **two digits: 0 and 1**.
2. **Bit:** The smallest unit of data in a computer is called a **bit** (binary digit). A bit is either 1 or 0.

3. **Writing:** Binary numbers are written as 110_2 or 10_2.
4. **Use:** They are mostly used in computers for programming and coding, because the computer understands only 0 and 1.
5. **MSB and LSB:** The bit on the extreme **left** is the **Most Significant Bit (MSB)**. The bit on the extreme **right** is the **Least Significant Bit (LSB)**.

Long Question: Differentiate between binary and decimal number systems. ***[Exercise (Brief Q2)]***

Binary Number System	Decimal Number System
It has base 2.	It has base 10.
It uses only two digits: 0 and 1.	It uses ten digits: 0 to 9.
It is used by computers.	It is used by people in daily life.
Each place is 2 times the place on its right (1, 2, 4, 8 ...).	Each place is 10 times the place on its right (1, 10, 100 ...).
It is written with base 2, like 1010_2.	It is written like 10_{10}, or without a base.
The leftmost bit is the MSB and the rightmost bit is the LSB.	It is also called the Hindu-Arabic number system.

Example: The binary number 1010_2 is equal to the decimal number 10_{10}.

Units of memory (Table 4.1)

Unit	Equal to
1 Bit	Binary Digit (0 or 1)
8 Bits	1 Byte (B)
1024 Bytes	1 Kilobyte (KB)
1024 KB	1 Megabyte (MB)
1024 MB	1 Gigabyte (GB)
1024 GB	1 Terabyte (TB)
1024 TB	1 Petabyte (PB)
1024 PB	1 Exabyte (EB)
1024 EB	1 Zettabyte (ZB)
1024 ZB	1 Yottabyte (YB)
1024 YB	1 Brontobyte
1024 Brontobyte	1 Geopbyte

Remember: In binary, 1 KB = 1024^1 bytes, 1 MB = 1024^2 bytes and so on. In the decimal way of counting, 1 KB is about 1000 bytes, 1 MB is about 1,000,000 bytes and 1 GB is about 1,000,000,000 bytes.

TOPIC 3: NUMBER SYSTEM CONVERSION: We can change a binary number into a decimal number, and a decimal number into a binary number.

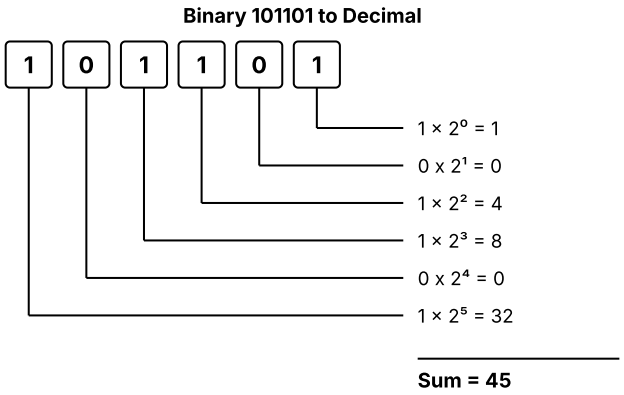
Long Question: How does a computer convert binary numbers into decimals? ***[Exercise (Brief Q3)]***

To change binary into decimal, we use the **positional notation method**. In this method, each digit has a **weight** based on its position. The weight is the base (2) raised to a power.

1. **Step 1:** Write the powers of 2 for all digits, starting from the **rightmost** digit. The first place is 2^0, the next is 2^1, then 2^2, and so on.
2. **Step 2:** Multiply each digit by its power of 2.
3. **Step 3:** Add all the results. The sum is the decimal number.

Example: Convert 101101_2 into decimal.

$1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 32 + 0 + 8 + 4 + 0 + 1 =$ **45**. So $\mathbf{(101101)_2 = (45)_{10}}$.



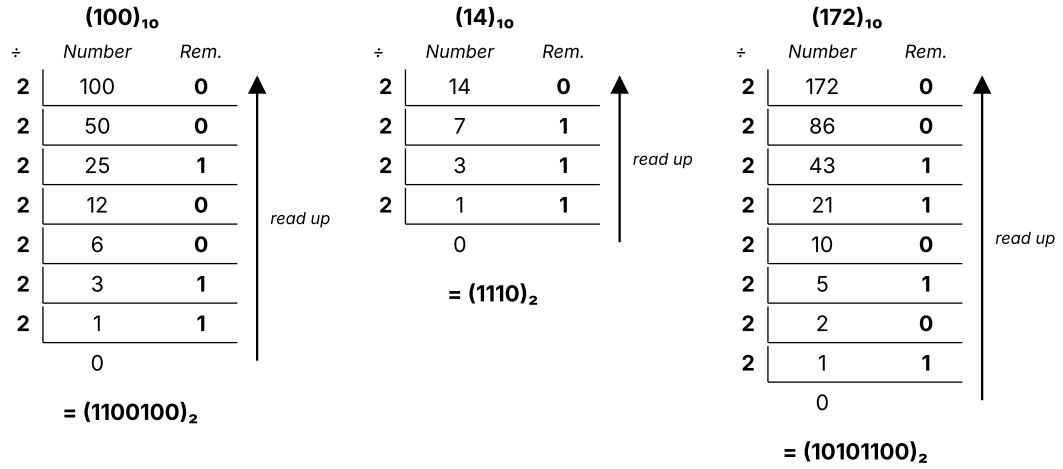
Binary to decimal using the positional notation method

Long Question: How does the computer convert decimal numbers into binary numbers? ***[Exercise (Brief Q4)]***

To change decimal into binary, we divide the number by 2 again and again. The base changes from 10 to 2.

1. **Step 1:** Divide the decimal number by **2** and write down the **remainder** (0 or 1).
2. **Step 2:** Divide the answer (quotient) by 2 again and write the remainder again.
3. **Step 3:** Repeat these steps until the quotient becomes **0**.
4. **Step 4:** Write the remainders in **reverse order**: the last remainder first. (The first remainder is the **LSB** and the last remainder is the **MSB**.)

Examples: $(100)_{10} = (1100100)_2$, $(14)_{10} = (1110)_2$ and $(172)_{10} = (10101100)_2$.



Decimal to binary: divide by 2 and read the remainders from bottom to top

Decimal to binary chart (0 to 20)

Decimal	Binary	Decimal	Binary	Decimal	Binary
0	0	7	111	14	1110
1	1	8	1000	15	1111
2	10	9	1001	16	10000
3	11	10	1010	17	10001
4	100	11	1011	18	10010
5	101	12	1100	19	10011
6	110	13	1101	20	10100

TOPIC 4: TEXT, IMAGE AND SOUND IN BINARY: Text, pictures and sound must also be changed into binary, so that the computer can store and process them.

Long Question: Write a note explaining the conversion of text, image and sound representation in binary code. ***[Exercise (Brief Q5)]***

1. Text:

- When we press a key on the keyboard, it is changed into a binary number. Then the computer processes it and the letter appears on the screen. *Example:* A = 0100 0001.

- We use a code to change each character into binary. One such code is **ASCII** (American Standard Code for Information Interchange). It gives a binary number to each character on the keyboard.
- ASCII is a **7-bit code**. It can store only **2^7 = 128 characters**. This is enough for English but not for other languages. It was extended to **2^8 = 256 characters**, but this was still not enough.
- So a new code, **Unicode**, was made. It can show **more than 65,000 characters**, so people of all languages (like Russian and Chinese) can use computers.

2. Images:

- Digital images are made up of small dots called **pixels**. Each pixel is stored as a binary number.
- In a black and white picture, **1 means black (on)** and **0 means white (off)**.
- The computer must know the size of the grid first. This data is called **metadata**. If the metadata is 10×10, the picture is 10 pixels across and 10 pixels down.
- Colours are stored as **8-bit numbers** using **RGB** (Red, Green, Blue).

3. Sound:

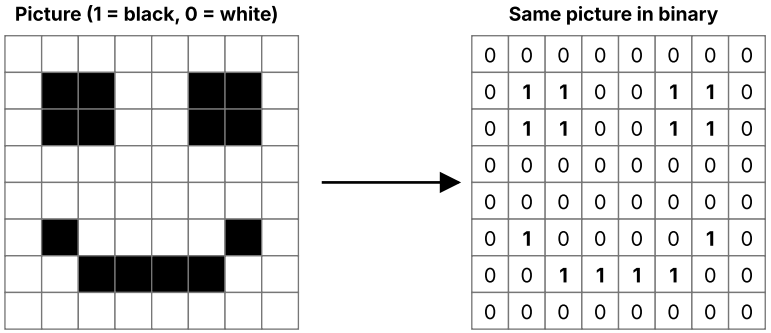
- Sound waves are **analogue**. To store them, we change the wave into numbers (binary). This is done by an **ADC (Analogue-to-Digital Converter)**.
- The ADC takes **samples** of the sound wave at regular intervals.
- The quality and size of the sound file depend on the **sample rate** and the **bit rate**.

Text examples (ASCII)

1. **Letter a:** 0110 0001 (decimal 97)
2. **Letter b:** 0110 0010 (decimal 98)
3. **Letter c:** 0110 0011 (decimal 99)

ASCII codes start at 0. They cover special characters like punctuation, the return (Enter) key and control characters, as well as number keys, capital letters and small letters.

Images and pixels



A simple black and white picture made with binary (1 = black, 0 = white)

Colours representation in binary code

1. **8-bit numbers:** Each colour part is stored with **8 bits**. This gives **256** levels of colour (0 to 255). *Example:* 0 = 00000000 and 255 = 11111111.
2. **RGB:** The most common method is to mix **Red, Green and Blue** light. Each part has a value from **0 to 255**. Computer screens add RGB light to make colours.
3. **Primary colours:** Red (255,0,0), Green (0,255,0) and Blue (0,0,255).
4. **Secondary colours:** Cyan (0,255,255), Magenta (255,0,255) and Yellow (255,255,0).
5. **Black and White:** Black is (0,0,0) and White is (255,255,255). Middle gray is (128,128,128).
6. **Hexadecimal:** Some programs show colours in **hexadecimal** numbers. Hexadecimal has **16 numerals**: 0 to 9 and A to F.

Colours in decimal, binary and hexadecimal

Colour	Decimal RGB	Binary RGB	Hexadecimal RGB
Black	(0,0,0)	(00000000, 00000000, 00000000)	(00,00,00)
White	(255,255,255)	(11111111, 11111111, 11111111)	(FF,FF,FF)
Red	(255,0,0)	(11111111, 00000000, 00000000)	(FF,00,00)
Green	(0,255,0)	(00000000, 11111111, 00000000)	(00,FF,00)
Blue	(0,0,255)	(00000000, 00000000, 11111111)	(00,00,FF)
Cyan	(0,255,255)	(00000000, 11111111, 11111111)	(00,FF,FF)
Magenta	(255,0,255)	(11111111, 00000000, 11111111)	(FF,00,FF)
Yellow	(255,255,0)	(11111111, 11111111, 00000000)	(FF,FF,00)
Gray	(128,128,128)	(10000000, 10000000, 10000000)	(80,80,80)
Pale Yellow	(200,180,120)	(11001000, 10110100, 01111000)	(C8,B4,78)

In the end, the screen (display) decides the colour we see. The same numbers, like (200,180,120), may not look exactly the same on all screens or printers. This makes accurate colour a difficult technical challenge.

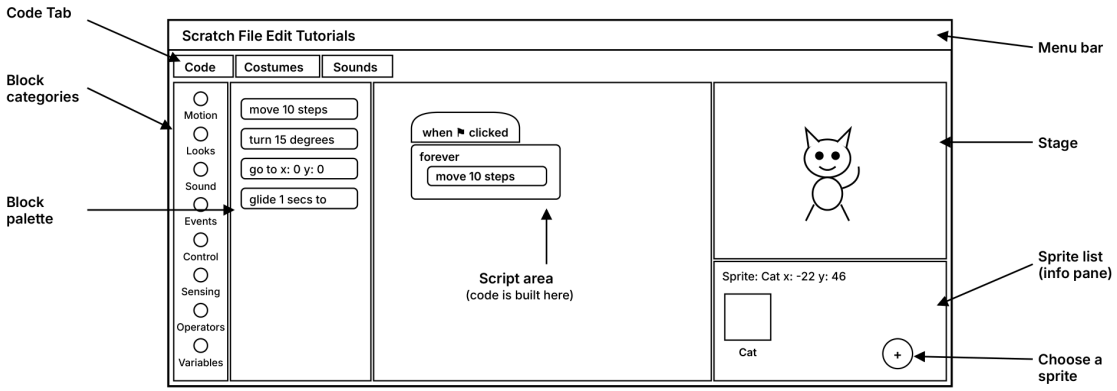
Sound

1. **Sample Rate:** It is the number of samples taken **every second**. More samples give **better sound quality**.
2. **Bit Rate:** It is the number of bits used to store **each sample**. More bits give **better accuracy**, but the **file size also becomes bigger**.
3. **Audacity:** It is a **free audio editor**. We can open an MP3 file in it and zoom in to see the value of each sample. We can also export the sound with different sample rates to hear the change in quality.

TOPIC 5: SCRATCH PROGRAMMING: Scratch is a programming software that makes it easy to create interactive games, animations and stories.

Scratch and its interface (4.4)

When Scratch opens, it shows one character by default. This character is called a **sprite** (the Scratch Cat). We can program the sprite to move and to interact with the user.



Scratch interface

Parts of the Scratch interface

Part	Description
Block Categories	Show the groups of blocks (Motion, Looks, Sound, Events, Control, Sensing, Operators, Variables). Click a category to see its blocks.
Block Palette	Has the list of blocks. Click a block and drag it to the script area.
Script Area	The place where we join blocks to make a program.
Stage	The place where the program runs and we see the sprite in action.
Sprite List	Shows the sprites used in the program. We can add, delete or change sprites and the backdrop (stage background) here.
X and Y Coordinates	Tell the exact position of the sprite on the stage. (0, 0) is the middle of the stage.

Other parts: Code tab (blocks palette), **Costumes tab** (paint editor), **Sounds tab** (sound editor), **Menu bar**, **Choose a Sprite** button, **Choose a Backdrop** button and **Extensions Library**.

Combination of programming constructs (4.4.1)

We combine **events** and **coordinates** so that the user can move a sprite, or the sprite can move by itself (automatically) without the user.

1. **Movement:** Movement is the most basic function in any game. It is the first thing we do in a new game. Just like we move our arms and legs in real life, players move their characters with the **arrow keys**. Movement lets characters interact more with the game world.

2. **Step 1, Select a Sprite:** Every object in a Scratch project is a sprite. Sprites are characters that move and run code. The Scratch Cat is in every project by default. To add a new sprite, click the **Choose a Sprite** button at the bottom right. It opens the **Sprite Menu**, a library of sprites. Click any character to add it.
3. **Step 2, Program the Sprite:** The easiest way to make a sprite move is to use **event listeners**. The code has a **yellow event block** (it tells the sprite **when** to move) and **blue motion blocks** (they actually **move** the sprite).

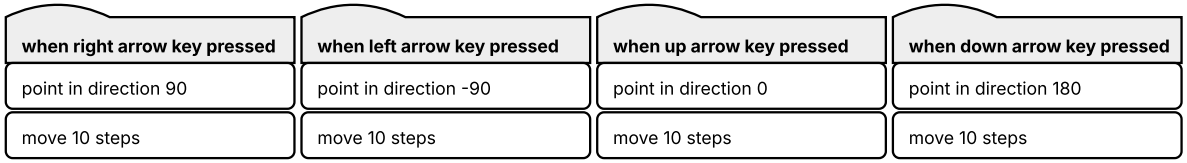
Tip: Right-click on code and click **Duplicate** to make a copy of the blocks. Then we do not need to drag and drop again. (The right-click menu also has Add Comment and Delete Block.)

Moving in four directions

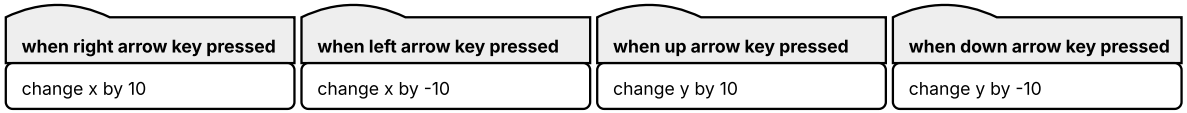
"When you press the right arrow key, point towards the right, then move forward 10 steps."

1. **Different keys:** Each script runs when a different arrow key is pressed.
2. **Different directions:** The "point in direction" block uses degrees: **90 = right, -90 = left, 0 = up and 180 = down**.
3. **Same speed:** The "move 10 steps" block is the same in all four scripts, so the sprite always moves at the same speed.

Step method: the sprite turns and moves



X-Y method: the sprite moves but does not turn



Moving a sprite with arrow keys: Step method and X-Y method

Long Question: Allow the user to move a sprite using arrow keys. *[Exercise (Project Q4)]*

There are three ways to make the sprite move right, left, up and down with the arrow keys:

1. **X-Y Method:** Select the sprite. Use **"when [arrow] key pressed"** with **"change x by 10"** (right), **"change x by -10"** (left), **"change y by 10"** (up) and **"change y by -10"** (down). The sprite moves, but it keeps pointing in the **same direction**.
2. **Step Method:** Use **"when [arrow] key pressed"**, then **"point in direction"** (90, -90, 0 or 180), then **"move 10 steps"**. The sprite **turns** while it moves. This is not good for a sprite that must turn for other reasons.
3. **Loop Method:** Both methods can also be put inside a **forever** loop with **"if key pressed"** blocks, so the program keeps checking the keys.

TOPIC 6: VARIABLES AND LOOPS IN SCRATCH: Variables help a program remember values. Loops repeat blocks again and again.

Variables

Sometimes a program must **remember** a value to use it many times. The best way is to use a **variable**. A variable is a named place that the computer uses to hold a value.

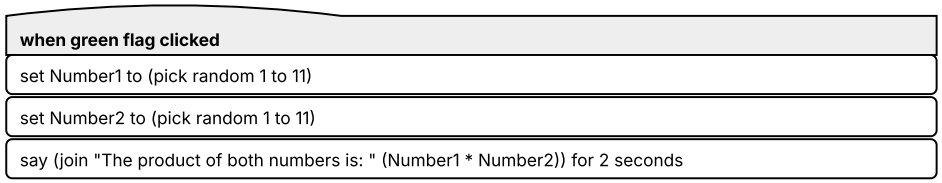
- We give a variable a **name** and call it by this name in the program.
- We assign a value to a variable. The value can be a **string, a number or a letter**.
- We can **change** the value of a variable many times.
- We can make **as many variables** as we want.

To create a variable in Scratch: Go to the **Variables** category and click **Make a Variable**. A "New Variable" box appears. Type a name, choose "For all sprites" or "For this sprite only" and click **OK**. Then blocks like set, change, show variable and hide variable appear.

1. **Set block: "set [variable] to (0)"** gives a value to the variable. Click the arrow to choose the variable. A variable can be set to a number (12) or a string (Hello). We should always set a value before using a variable.
2. **Reporter block:** The round **(variable)** block reports (gives) the value saved in the variable. *Example:* "say (message1) for 2 seconds" shows the value of message1.

Creating, setting and changing a variable under conditions

The **pick random** operator gives a random number in a range. In the script below, Number1 and Number2 get random numbers from 1 to 11, and the sprite says their product. To show the two random numbers as well as the product, we use more **nested join** blocks.



Using variables and pick random to show the product of two numbers

Working with loops in Scratch

The **Control** category has two types of loop blocks:

1. **Repeat Loop:** It repeats the blocks inside it for a **fixed number of times** that we write at the start.
2. **Forever Loop:** It keeps repeating the blocks inside it until a condition stops it. If we do not use a condition, it becomes an **infinite loop**.

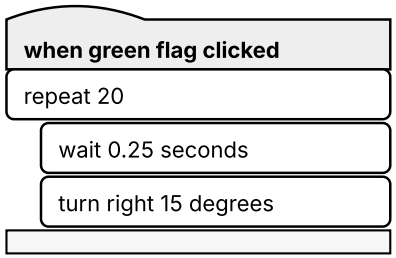
Program 1: Change the direction of the sprite by 15 degrees, 20 times, with a delay of 0.25 seconds. **Solution:** when green flag clicked, repeat 20 [wait 0.25 seconds, turn 15 degrees].

Long Question: Create a countdown timer in Scratch. ***[Exercise (Learning outcome, Program 2)]***

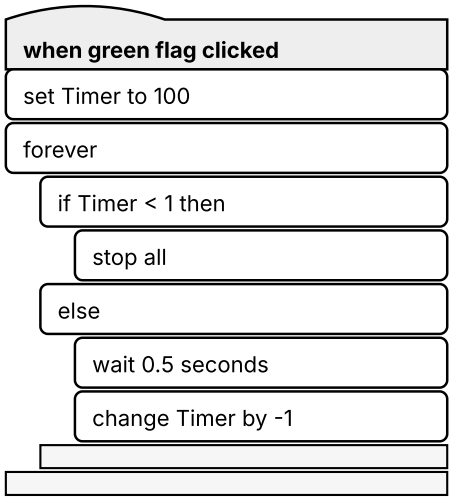
Program: Display a timer that starts from 100 and stops the program at 0. The value should change every 0.5 seconds.

1. Make a variable named **Timer**.
2. Use **"when green flag clicked"** to start the program.
3. Use **"set Timer to 100"** to give the starting value.
4. Put a **forever** loop.
5. Inside it, use **"if Timer < 1 then"**, **"stop all"**. This stops the program at 0.
6. In the **else** part, use **"wait 0.5 seconds"** and **"change Timer by -1"**. The timer goes down by 1 every half second.

Repeat loop: turn 15° 20 times



Forever loop: countdown timer

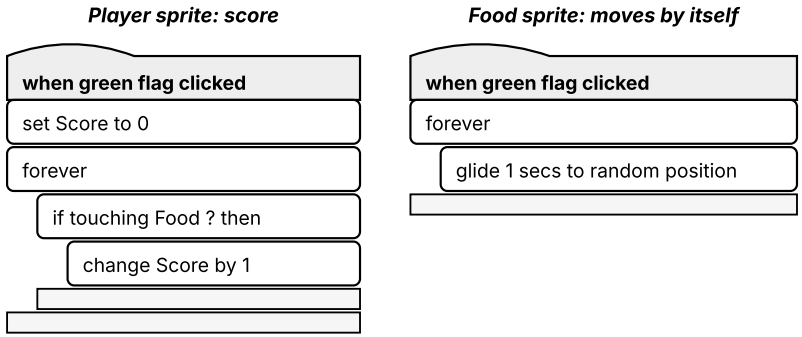


Repeat loop and forever loop (countdown timer)

Long Question: Create a game with a player sprite and a food sprite using coordinates, conditionals, loops and variables. ***[Exercise (Project Q3, Q5, Q6)]***

1. **Two sprites:** Add a **player** sprite and a **"food"** sprite that the player has to chase.
2. **Move the player (events + coordinates):** Use "when [arrow] key pressed" with "change x by" or "change y by" blocks, so the player moves right, left, up and down with the arrow keys.
3. **Move the food automatically:** In the food sprite, use "when green flag clicked", "forever" and **"glide 1 secs to random position"** (or "go to random position"). The food moves by itself, without the user.

4. **Score variable:** Make a variable **Score**. Set it to **0** when the game starts.
5. **Conditional:** Inside a forever loop, use **"if touching Food? then change Score by 1"**. The player gets a point when it touches the food. (We can also give a point when the player clicks on the moving sprite.)



Scripts for the food-chasing game

SHORT QUESTIONS — UNIT 4

1. Describe how computers save information using binary codes. ***[Exercise]***

Computers store all information as 0s and 1s. Tiny switches inside the computer are off (0) or on (1). Each switch is a bit, and 8 bits make a byte.

2. Differentiate between binary and decimal number systems. ***[Exercise]***

The binary system has base 2 and uses 0 and 1. The decimal system has base 10 and uses 0 to 9. Computers use binary and people use decimal.

3. How does a computer convert binary numbers into decimals? ***[Exercise]***

Multiply each binary digit by its power of 2 (2^0, 2^1, 2^2 … from the right) and add all the results. *Example:* $101101_2 = 45$.

4. How does a computer convert decimal numbers into binary? ***[Exercise]***

Divide the number by 2 again and again and note the remainders until the quotient is 0. Then write the remainders from last to first. *Example:* $14 = 1110_2$.

5. Write a note on text representation in binary. ***[Exercise]***

Each key is changed into a binary code using ASCII or Unicode. *Example:* A = 0100 0001. ASCII stores 128 characters and Unicode stores more than 65,000.

6. What is a bit? ***[Book concept]***

A bit (binary digit) is the smallest unit of data. It is either 0 or 1.

7. What is a byte? ***[Book concept]***

Eight bits together make one byte.

8. Who refined the binary number system? ***[Book concept]***

Gottfried Leibniz refined the binary system in the 17th century. He invented it around 1679 and published it in 1701.

9. What are MSB and LSB? ***[Book concept]***

In a binary number, the leftmost bit is the Most Significant Bit (MSB) and the rightmost bit is the Least Significant Bit (LSB).

10. What is ASCII? Write its full form. ***[Book concept]***

ASCII (American Standard Code for Information Interchange) is a 7-bit code that gives a binary number to each character on the keyboard.

11. Why was Unicode created? ***[Book concept]***

ASCII could store only 128 (later 256) characters, which was not enough for other languages. Unicode can show more than 65,000 characters.

12. What are pixels? ***[Book concept]***

Digital images are made of small dots called pixels. Each pixel is stored as a binary number.

13. What is metadata of an image? ***[Book concept]***

Metadata is the data that tells the size of the image grid, like 10×10 pixels.

14. What is RGB? ***[Book concept]***

RGB means Red, Green and Blue. Colours are made by mixing these three lights, each with a value from 0 to 255.

15. What is hexadecimal? ***[Book concept]***

It is a number system with 16 numerals: 0 to 9 and A to F. Some programs show colours in it, like FF,00,00 for red.

16. What is an ADC? ***[Book concept]***

ADC (Analogue-to-Digital Converter) changes analogue sound waves into numbers by taking samples at regular intervals.

17. Differentiate between sample rate and bit rate. ***[Book concept]***

Sample rate is the number of samples taken every second. Bit rate is the number of bits used to store each sample.

18. What is a sprite? ***[Book concept]***

A sprite is a character or object in Scratch that can move and run code. The Scratch Cat is the default sprite.

19. What is a variable? ***[Book concept]***

A variable is a named place that holds a value, such as a number, a letter or a string. Its value can be changed.

20. Differentiate between repeat and forever loop. ***[Book concept]***

A repeat loop runs a fixed number of times. A forever loop keeps running until a condition stops it.

21. What do the numbers 90, -90, 0 and 180 mean in the point in direction block? ***[Book concept]***

They are degrees: 90 means right, -90 means left, 0 means up and 180 means down.

22. What is the use of the Duplicate option in Scratch? ***[Book concept]***

Right-click on code and choose Duplicate to make a copy of the blocks, so we do not have to drag them again.

MULTIPLE CHOICE QUESTIONS — UNIT 4

1. The term __ also refers to any digital encoding/decoding system in which there are exactly two possible states. ***[Exercise MCQ 1]***

a) Digital ✔ **b) Binary** c) Python d) Programming

2. __ bits together make what we call a "byte" of the data. ***[Exercise MCQ 2]***

a) 6 b) 7 ✔ **c) 8** d) 9

3. To convert numbers from decimals to binary, the given decimal number is divided repeatedly by __. ***[Exercise MCQ 3]***

✔
a) 2 b) 3 c) 4 d) 5

4. __ are made up of pixels. ***[Exercise MCQ 4]***

a) Binary numbers b) Decimals c) Programs ✔ **d) Digital images**

5. ASCII code can only store up to __ characters. ***[Exercise MCQ 5]***

✔
a) 128 b) 256 c) 64 d) 124

6. __ are represented in computer using 8-bit numbers. ***[Exercise MCQ 6]***

a) Numbers b) Letters c) Images ✔ **d) Colors**

7. We need __ to convert the sound waves into numerical representation. ***[Exercise MCQ 7]***

a) Audio to digital converter b) Audio to digits converter ✔ **c) Analogue to digital converter** d) Analogue to digits converter

8. The best way to remember a value when designing the program is to use a __. ***[Exercise MCQ 8]***

✔
a) Variable b) Value c) Number d) Binary

9. The __ is/are analogue. ***[Exercise MCQ 9]***

a) Light b) Numbers c) Values ✔ **d) Sound**

10. The decimal number system represents a number with a base of __. ***[Exercise MCQ 10]***

a) 2 b) 3 c) 5 ✔ **d) 10**

11. Bit is the short form of:

a) Binary Item ✔ **b) Binary Digit** c) Basic Digit d) Byte Digit

12. The smallest unit of data in a computer is:

a) byte ✔ **b) bit** c) nibble d) kilobyte

13. The binary number system was refined by:

a) Charles Babbage ✔ **b) Gottfried Leibniz** c) Bill Gates d) Isaac Newton

14. In a computer, a closed (on) switch represents:

a) 0 ✔ **b) 1** c) 2 d) 10

15. 1024 Kilobytes make one:

a) Gigabyte ✔ **b) Megabyte** c) Terabyte d) Byte

16. The binary number system has base:

✔
a) 2 b) 8 c) 10 d) 16

17. The decimal number system is also called the __ number system.

a) Roman ✔ **b) Hindu-Arabic** c) Greek d) Binary

18. The leftmost bit in a binary number is called:

a) LSB ✔ **b) MSB** c) Byte d) Pixel

19. $(101101)_2$ is equal to:

a) 40 ✔ **b) 45** c) 54 d) 31

20. $(14)_{10}$ in binary is:

a) 1101 b) 1011 ✔ **c) 1110** d) 1111

21. $(1010)_2$ is equal to decimal:

a) 8 ✔ **b) 10** c) 12 d) 5

22. ASCII is a __ bit code.

✔
a) 7 b) 8 c) 16 d) 32

23. In ASCII, the letter "a" has the decimal value:

a) 65 ✔ **b) 97** c) 98 d) 99

24. Unicode can represent more than __ characters.

a) 128 b) 256 ✔ **c) 65,000** d) 1,000

25. In a black and white picture, 1 means:

a) white ✔ **b) black** c) red d) blue

26. The data that tells the size of an image is called:

a) pixel ✔ **b) metadata** c) bit rate d) sample

27. In RGB, white is represented by:

a) (0,0,0) ✔ **b) (255,255,255)** c) (255,0,0) d) (128,128,128)

28. Hexadecimal has __ numerals.

a) 2 b) 10 ✔ **c) 16** d) 8

29. The number of samples taken every second is called:

a) bit rate ✔ **b) sample rate** c) metadata d) pixel

30. Audacity is a free:

a) image editor ✔ **b) audio editor** c) web browser d) game

31. In Scratch, the default character is called a:

a) pixel ✔ **b) sprite** c) stage d) block

32. In Scratch, the location (0, 0) is in the __ of the stage.

a) top b) bottom ✔ **c) middle** d) left corner

33. In "point in direction", 180 means:

a) right b) left c) up ✔ **d) down**

34. The loop that runs a fixed number of times is:

a) forever ✔ **b) repeat** c) infinite d) if

35. A forever loop without any condition becomes a/an __ loop.

a) finite ✔ **b) infinite** c) repeat d) nested

Mistakes Found in the Book

1. Page 61: The picture says $(172)_{10} = (1010100)_2$. This is wrong, because $1010100_2 = 84$. The correct answer is **$(172)_{10} = (10101100)_2$**.
2. Page 72: The text says the script shows "the product" of both numbers, but the picture puts Number1 and Number2 side by side with no **x (multiply)** operator between them.
3. Page 73: In the arrow-key scripts, many blocks say "up arrow" for every direction, and the step method says "move 10 **stage**" instead of "move 10 **steps**". The correct scripts are shown in these notes.
4. Page 74: Under "Repeat Loop", the book shows a picture of the **forever** block instead of the repeat block.
5. Page 57: It says a gigabyte is "approximately trillions of bytes". In fact, a gigabyte is about one **billion** bytes.
6. Page 58, Table 4.1: "Terra Byte" should be "**Tera**byte".
7. Page 78, MCQ 3: It says "till we get 0 or 1 as the final quotient". The book's own method (page 61) divides until the quotient is **0**.

Computer Science — Class 7

Single National Curriculum (SNC) — Punjab Curriculum and Textbook Board

Unit 5 — Digital Citizenship

TOPIC 1: ETHICS AND ETHICAL ISSUES: Ethics tells us what is right and what is wrong. It guides how we should behave with other people.

Ethics (5.1)

Ethics is the subject that deals with what is **morally good and bad**, or **morally right and wrong**. The word is also used for any system of **moral values or principles**. Ethics is based on clear standards of right and wrong. These standards tell people what they should do, in terms of rights, duties (obligations), benefits to society, fairness and good qualities (virtues).

Extra Bit: Aristotle said that ethics is the study of human relations in their most perfect form. He called it **the science of proper behaviour**.



Ethics for success

Ethical issues

An ethical issue is a situation in which a **moral conflict** arises in a society or a workplace. In it, a moral standard is being challenged. Ethical issues include:

1. Privacy and confidentiality
2. Issues related to socially vulnerable (weak) people

3. Health insurance discrimination
4. Employment (job) discrimination
5. Individual responsibility
6. Issues related to race and ethnicity
7. Implementation of laws

TOPIC 2: ETHICS IN DIGITAL ENVIRONMENT: Today most of our relationships have moved online, to chats, messengers, social media and other ways of online communication.

Long Question: Define ethics in digital environment. ***[Exercise (Detail Q1)]***

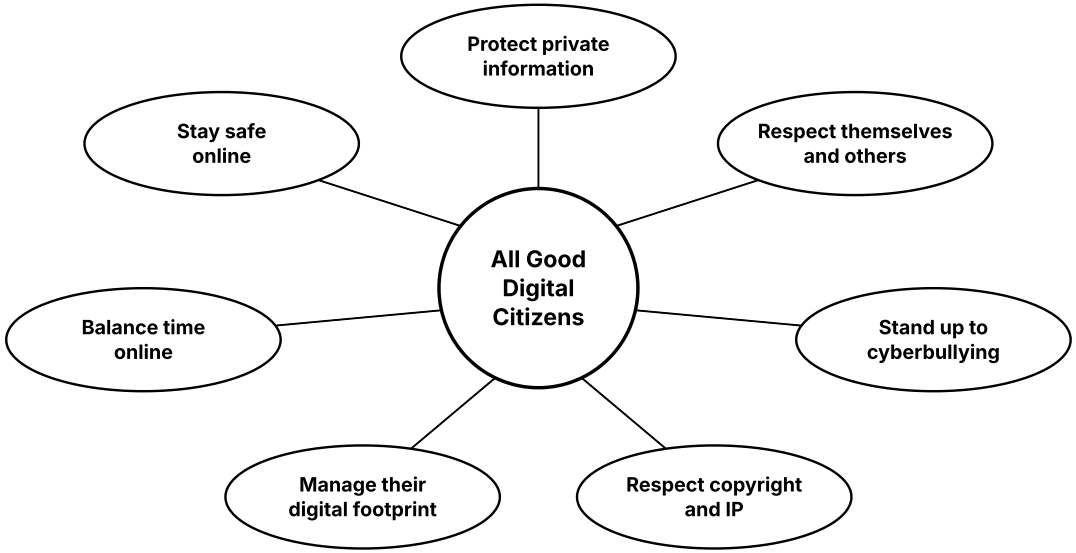
1. **Meaning:** Today we live in a digital world. Most of our relationships have moved online to chats, messengers and social media. Knowing and obeying the rules (principles) of good behaviour while being online is called **digital ethics**.
2. **Benefits of technology:** Technology has given us many benefits, such as **5G, Virtual Reality, Mixed Reality, Artificial Intelligence, driverless vehicles, digital assistants, robots, smart devices and automation**.
3. **Negative impacts:** Technology also has bad effects. Some common ethical issues in the digital environment are:

- Misuse of personal information
- Misinformation (wrong information) and deep fakes
- Lack of oversight and not accepting responsibility
- Social and political instability

1. **Good behaviour:** We should respect others online, protect privacy, not copy others' work and not spread false news.

Long Question: Discuss the importance of being safe and responsible digital citizenship. ***[Exercise (Detail Q2)]***

1. **Why it is important:** We spend most of our time online and store most of our things online. So it is very important to protect our **devices and information** from harmful (malicious) people.
2. **Danger of being unsafe:** If we leave our devices unprotected, the result can be small, like a **slower computer**, or very big, like **losing all the money in a bank account** or **identity theft**.
3. **Good digital citizen:** A good digital citizen encourages positive and healthy talk online. He or she knows about common online scams and toxic behaviour, avoids **cyberbullying** and shows **empathy** (care) for others online.
4. **Characteristics of all good digital citizens:** They protect private information for themselves and others, respect themselves and others, stay safe online, stand up to cyberbullying, balance the time they spend online, respect copyright and intellectual property, and carefully manage their digital footprint.



Characteristics of a good digital citizen

<u>Long Question: What do you mean by responsible digital citizen?</u> ***[Exercise (Detail Q8)]***

1. **Responsible Digital Citizenship:** We must understand the results of posting photos and videos and uploading other personal content. Once something is online, it is **very hard to remove**. It can become part of our **permanent online reputation**. Photos can also be changed or shared without our permission.
2. **Respectful Digital Citizen:** Respect for ourselves and others is important in all relationships, and it is the same online. We should treat online friends with as much respect as face-to-face friends. We should **not create or forward** nasty or humiliating emails, pictures or messages about anyone.
3. **Online social skills:** A responsible digital citizen has the online social skills to take part in online community life in an **ethical and respectful** way.
4. Properties of a good digital citizen:

- Respect the rights of others.
- Practise tolerance.
- Be informed about the world around you.
- Respect the property of others.
- Be compassionate (kind).
- Take responsibility for your actions.
- Pay taxes.

Extra Bit: If you get nasty or bullying comments, you should **block or unfriend** people who do not treat you with respect.

Long Question: How can we protect our reputation online? ***[Exercise (Detail Q3)]***

Being a responsible digital citizen means having the online social skills to take part in online community life in an ethical and respectful way. To protect our reputation, responsible digital citizenship means:

1. **Behaving lawfully:** Follow the law when we are online.
2. **Protecting privacy:** Protect our own privacy and the privacy of others.
3. **Knowing rights and responsibilities:** Know our rights and responsibilities when we use digital media.
4. **Thinking before acting:** Think about how our online activities affect **ourselves, people we know and the wider online community**.
5. **Careful posting:** Think before posting photos and videos, because once they are online they are hard to remove and become part of our permanent online reputation.

Protect your privacy

1. Share only as much personal information as necessary.
2. Keep privacy settings up to date on social media sites, so that your profile is not open to everyone.
3. Keep passwords private.
4. Check the location settings and services on smartphones, tablets and apps.

Keep an eye out for scams

Cybercriminals can trick (scam) people online in many ways. Most scams can be avoided with the right knowledge and tools. Three common scams are:

1. **Phishing:** A scammer sends a **malicious (harmful) link**. It is made to trick a person into giving **sensitive information** to the attacker. The attacker can then watch everything the victim does on the site.
2. **Fake Websites:** These are common with **shopping websites**. Cybercriminals make a fake site to make you enter your personal information. Then they misuse it.
3. **Tech Support Scams:** Fake tech support agents ask for **login information or remote access** to your computer. Then they ask for money to fix it.

Long Question: How can we say that Tech Support scams are harmful? ***[Exercise (Detail Q7)]***

1. **Fake agents:** In a tech support scam, a person pretends to be a computer expert (tech support agent). Often a **fake warning** appears on the screen, such as "System may have detected viruses on your computer. Call tech support right away."
2. **Stealing information:** The fake agent asks for our **login information** (user name and password). With it, he can open our accounts.
3. **Remote access:** He asks for **remote access** to our computer. Then he can see our files, steal our data or install harmful software.
4. **Asking for money:** After that, he asks for **money** to "fix" the computer. This is a **red flag** (warning sign). A real tech support agent would not usually ask for any payment like this.

Conclusion: So tech support scams are harmful because we can lose our private data, our money and the safety of our computer. We should never give passwords or remote access to unknown people.

TOPIC 3: USES OF INTERNET: Good use of the Internet makes our lives easy and simple. The Internet gives us useful data, information and knowledge for personal, social and economic development.

Use of Internet in business (5.3.1)

The Internet has become a very important (integral) part of business. Its uses in business are much more than email and a basic website. Common uses are:

1. **E-commerce (5.3.2):** According to the U.S. Census Bureau, e-commerce was **14 percent of all retail sales in 2020**. Government services are also available online. Businesses like **Amazon** and **Shopify** stores do all their sales and customer dealings on the Internet.
2. **Marketing and CRM (5.3.3): CRM (Client Relationship Management)** is software that uses a wide range of data to help businesses serve customers better, increase sales and find new customers. CRM analytics lets a business treat each customer in a personal way. Tools like **Facebook Pixel** can find new customers based on the websites they visit, the posts they like and the videos they watch on YouTube.
3. **Collaboration and Document Storage (5.3.4):** Tools like **Google Drive, Microsoft OneDrive and Dropbox** let people work together on projects in real time from anywhere. They can update, edit and comment on documents. The creator of the file controls who can see and edit it.
4. **Videoconferencing, Chat and Remote Employees (5.3.5):** Many businesses let people work from home (remote work). This saves the cost of offices. They use **video conferencing, chat applications and social media** to keep teams together. **Automated chatbots** on websites answer customer questions quickly.



CRM: from leads to customers

Use of Internet in social networking (5.3.6)

Social networking means using Internet-based social media sites to **stay connected** with friends, family, colleagues or customers. It can have a social purpose, a business purpose or both. *Examples:* Facebook, Twitter, Instagram and Pinterest. It also means making and keeping personal and business relationships using technology.

Use of Internet in entertainment (5.3.7)

We can enjoy a lot of content on the Internet through our phone, TV or computer. Popular free online entertainment resources are:

- YouTube videos
- Podcasts
- Music streaming, such as Pandora or Spotify
- News sites, such as ARY, Geo, Bol News, Dunya News and National Public Radio
- Game sites, such as King.com and Miniclip

TOPIC 4: COPYRIGHT, PLAGIARISM AND PIRACY: We must respect the work of other people. Copying it without permission is wrong and can be a crime.

Copyright (5.4)

Copyright is a type of **intellectual property right** that protects the **original works of an owner**. Under copyright law, works include paintings, photographs, illustrations, musical compositions, sound recordings, computer programs, books, poems, blog posts, movies, architectural works and plays.

Plagiarism (5.5)

Presenting **someone else's work or ideas as your own**, with or without their consent, is called plagiarism. It covers all published and unpublished material, in electronic or printed form. Plagiarism is **academic dishonesty** and a **disciplinary offence** (punishable act).

Types of plagiarism

1. **Global Plagiarism:** Copying (plagiarizing) a **whole text**.
2. **Paraphrasing Plagiarism:** Rewriting someone else's ideas in new words and showing them as your own thoughts.
3. **Patchwork Plagiarism:** Copying phrases and ideas from **different sources** and joining them into a new text.
4. **Self-Plagiarism:** Using again (recycling) your **own old work** that you have already submitted or published.

Piracy (5.6)

The act of **illegally copying (reproducing) copyrighted material**, such as books, computer programs and films, is called piracy. It is a **criminal act** with the aim of stealing other people's ideas and inventions.

Difference between piracy and plagiarism

Plagiarism	Piracy
Showing someone else's work or ideas as your own	Illegally copying and using or selling copyrighted material
The person takes credit for another person's work	The person copies the work without permission or payment
It is academic dishonesty and a disciplinary offence	It is a crime (criminal act)
***Example:* copying an essay from the Internet and putting your name on it**	*Example:* copying a movie or software and selling it

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TOPIC 5: DIGITAL MEDIA BIAS: Media bias happens when a news channel allows its own opinions to affect the news it reports.

Digital media bias (5.7)

Media criticism means examining and studying the messages in mass media. A media channel may show bias in **how** it reports a news story, or in **which stories** it chooses to cover, because it thinks some stories are more important than others.

Types of media bias

1. Slant Bias
2. Spin Bias
3. Sensationalism Bias
4. Story Choice Bias
5. Word Choice Bias
6. Omission Bias

Long Question: Discuss the impacts of Media Bias. ***[Exercise (Detail Q6)]***

1. **One-sided news:** Media bias shows up in the **selection of news stories** (what editors think is most important) and in **how** they are covered. So people get only one side of the story.
2. **Wrong opinions:** When a news channel lets its opinions affect the news, people may form wrong opinions about people, groups or events.
3. **Hidden information:** Because of **omission bias** and **story choice bias**, some important news is not shown at all.
4. **Exaggeration: Sensationalism bias** makes small news look very big and exciting, which can create fear or anger.
5. **Hiding criticism:** Calling a media source "unbiased" can also be a way to **hide bias** or to stop criticism, debate and dialogue.

Importance of unbiased media (5.8): Unbiased news is a good thing. News channels should try to give a **balanced view**, collect views from all sides of politics, know about different worldviews and avoid media bias. Unbiased news fairly shows many points of view on issues on which people disagree. Biased news from one side (such as a conservative view) is also part of media, as long as the channel **openly shows its bias**.

Long Question: How can we avoid Media Bias? ***[Exercise (Detail Q4)]***

We can do the following things to avoid the problems of media bias:

1. Provide multiple perspectives (views) on news and issues to empower the reader.
2. Strengthen our **media literacy** skills.
3. Try to get **all sides** of an issue.
4. Watch **different news channels** with different tendencies (views).
5. **Remember:** if it sounds **too good to be true**, it probably is not true.
6. Read the **footnotes and caveats** (small warnings) in advertisements or promotions.

TOPIC 6: MISUSE OF COMPUTER RESOURCES AND PRIVACY:

Computer resources must be used in the right way. We must also keep our personal information private and safe.

Long Question: Enlist the misuse of computer resources. ***[Exercise (Detail Q5)]***

Misuse of computer resources is not only physical misuse. It also includes:

1. **Unauthorized Access:** Using computers, computer software, computer data or networks **without permission (authorization)**.
2. **Improper Use:** Using computer resources for a purpose **other than** the purpose for which they were allowed.
3. **Illegal Use:** Breaking (violating) any **software licence or copyright**.
4. **Interfering with Others: Harassing or threatening** other users, or disturbing their access to SOA (Service Oriented Architecture) computing facilities.
5. **Improper Alteration of System Files:** Changing accounting system files or audit trails **without permission**, to change or delete records of use.

How to secure information privacy and confidentiality (5.10)

1. **Privacy:** It is a person's **right to control** how his or her personal information is collected, used and shared.
2. **Confidentiality:** It means **how much** information can be kept **secret**.

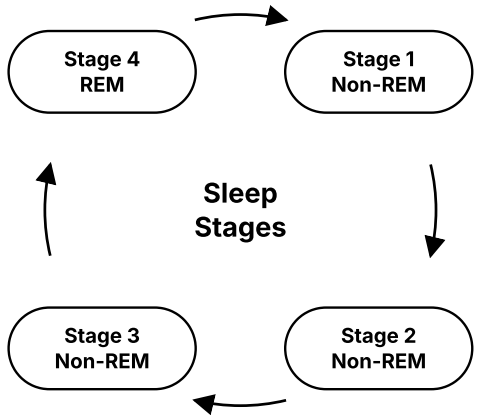
Not knowing how to protect our personal information online can cost us a lot. It can also cause stress, or worse, if wrong people get our personal data. Important tips are:

1. **Pick strong and unique passwords:** Do not use easy passwords linked to your personal life. Mix **capital letters, numbers and symbols**, and use different passwords for different accounts.

2. **Use a VPN:** A **VPN (Virtual Private Network)** adds an extra layer of privacy. It **encrypts** the data that leaves your computer and sends it through third-party servers.
3. **Avoid clicking links and attachments in emails:** Phishing emails tell sad stories or have strange links to make us give our details. **Scan attachments** with anti-malware software first.
4. **Log out of websites properly:** If we do not log out, we leave the door open for others. It is like leaving your wallet in a shop. If a laptop with automatic logins is stolen, the thief can easily open private data. Use a **password manager** instead of auto logins.
5. **Use your browser's incognito mode:** The next person cannot see your **browsing history**. It also deletes **cookies and temporary files** that track your online activity.
6. **Keep your anti-malware up to date: Trojan horses and worms** can infect a computer and send whatever you type to criminals. Updated anti-malware stops them.
7. **Be careful what you share online:** Instagram and Twitter are hotspots for phishers and cybercriminals. Do not publicly share your **identity, location or date of birth**.

TOPIC 7: POSSIBLE DANGERS OF THE INTERNET: The Internet is a great place to find information, but it is also full of dangers. Many users do not notice these dangers.

1. **Depression and Anxiety:** Spending too much time on social networking sites can badly affect our **mood**. People who use social media too much are more likely to have poor mental health, including **anxiety and depression**.
2. **Cyberbullying:** Bullying people by using **electronic means** is called cyberbullying. It is also called **online bullying**. It is common among teenagers. *Example:* when someone bullies or harasses others on the Internet, especially on social media sites.
3. **FOMO (Fear of Missing Out):** It became common with the rise of social media. It is one of the most widespread **negative effects** of social media. People keep posting online only because they fear that others will think their life is not wonderful or important.
4. **Unrealistic Expectations:** Social media creates **unrealistic (false) expectations** of life and friendship in our minds. Most social media lacks honesty (authenticity). People share only exciting adventures and heavily staged (posed) photos.
5. **Unhealthy Sleep Patterns:** Spending too much time on social media can cause **poor sleep**. If your sleep becomes irregular and your work suffers, reduce the time you spend on social media.



Sleep stages

SHORT QUESTIONS — UNIT 5

1. Define ethics. ***[Exercise]***

Ethics is the subject that deals with what is morally good and bad, or morally right and wrong.

2. Justify that plagiarism is an offence. ***[Exercise]***

Plagiarism means showing someone else's work or ideas as your own. It is cheating and academic dishonesty, so it is a disciplinary offence.

3. Define phishing in your words. ***[Exercise]***

In phishing, a scammer sends a harmful link to trick a person into giving sensitive information like passwords or bank details.

4. Enlist issues of digital ethics. ***[Exercise]***

Misuse of personal information, misinformation and deep fakes, lack of oversight and responsibility, and social and political instability.

5. Discuss any three uses of internet in business. ***[Exercise]***

E-commerce (selling online, like Amazon), marketing and CRM (serving customers better), and collaboration and document storage (Google Drive, OneDrive, Dropbox).

6. Define CRM. ***[Exercise]***

CRM (Client Relationship Management) is software that uses data to help businesses serve customers better, increase sales and find new customers.

7. Discuss use of internet in entertainment. ***[Exercise]***

We enjoy YouTube videos, podcasts, music streaming (Spotify), news sites (ARY, Geo) and game sites (Miniclip) on our phone, TV or computer.

8. Define copyright in your words. ***[Exercise]***

Copyright is a legal right that protects the original work of an owner, like books, songs, movies and computer programs.

9. Differentiate between piracy and plagiarism. ***[Exercise]***

Plagiarism is showing someone else's work as your own. Piracy is illegally copying copyrighted material like books, software and films.

10. What is an ethical issue? ***[Book concept]***

It is a situation in which a moral conflict arises in a society or workplace.

11. What is digital ethics? ***[Book concept]***

Knowing and obeying the principles of good behaviour while being online is called digital ethics.

12. What is a fake website scam? ***[Book concept]***

Cybercriminals make a fake website, mostly a shopping site, to get our personal information and misuse it.

13. What is social networking? ***[Book concept]***

It is the use of Internet-based social media sites to stay connected with friends, family, colleagues or customers.

14. Name the types of plagiarism. ***[Book concept]***

Global, paraphrasing, patchwork and self-plagiarism.

15. What is self-plagiarism? ***[Book concept]***

Using your own old work again, which you have already submitted or published.

16. What is media bias? ***[Book concept]***

Media bias happens when a news channel allows its opinions to affect the news it reports.

17. Name the types of media bias. ***[Book concept]***

Slant, spin, sensationalism, story choice, word choice and omission bias.

18. Differentiate between privacy and confidentiality. ***[Book concept]***

Privacy is a person's right to control how personal information is used and shared. Confidentiality is how much information can be kept secret.

19. What is a VPN? ***[Book concept]***

A VPN (Virtual Private Network) encrypts our data and sends it through other servers to give extra privacy online.

20. What is incognito mode? ***[Book concept]***

It is a browser mode that does not save browsing history and deletes cookies and temporary files.

21. What is cyberbullying? ***[Book concept]***

Bullying or harassing others using electronic means, like social media, is called cyberbullying.

22. What is FOMO? ***[Book concept]***

FOMO means Fear of Missing Out. People keep posting online because they fear others will think their life is not wonderful.

23. What is unauthorized access? ***[Book concept]***

Using computers, software, data or networks without permission is called unauthorized access.

24. What should we do if we get bullying comments? ***[Book concept]***

We should block or unfriend the people who do not treat us with respect.

MULTIPLE CHOICE QUESTIONS — UNIT 5

1. The ways of online communication involve: ***[Exercise MCQ 1]***

a) Social media b) Messengers c) chats **d) All** ✔

2. Leaving your devices unprotected can result in: ***[Exercise MCQ 2]***

a) Slow down your computer ✔ b) Fast your computer c) Gain money d) All

3. In __ scammer sends you a malicious link, designed to trick a person into revealing sensitive information. ***[Exercise MCQ 3]***

a) Fake Websites	b) Tech Support	✔ **c) Phishing**	d) None

4. Fake __ agents will often ask for login information or remote access to your computer. ***[Exercise MCQ 4]***

a) Fake Websites	✔ **b) Tech Support**	c) Phishing	d) None

5. Which of the following is not a use of internet? ***[Exercise MCQ 5]***

a) E-commerce	b) CRM	✔ **c) SRM**	d) All

6. CRM stands for: ***[Exercise MCQ 6]***

a) Client Relationship Model	b) Customer Relation Module	✔ **c) Client Relationship Management**	d) Client Relation Module

7. The term __ refers to the use of internet-based social media sites. ***[Exercise MCQ 7]***

✔ **a) social networking**	b) social websiting	c) media networking	d) websiting

8. __ is a type of intellectual property rights that protects original works of an owner. ***[Exercise MCQ 8]***

a) cyberbullying	b) plagiarism	c) piracy	✔ **d) copyright**

9. Illegally reproducing copyrighted material, such as books, computer programs and films, is called: ***[Exercise MCQ 9]***

a) cyberbullying	b) plagiarism	✔ **c) piracy**	d) copyright

10. Accessing computers, software, data or networks without authorization is called: ***[Exercise MCQ 10]***

✔ **a) unauthorized access**	b) illegal use	c) authorized access	d) improper experiments

11. Ethics deals with what is morally:

a) fast and slow	✔ **b) right and wrong**	c) big and small	d) old and new

12. Aristotle called ethics the science of:

a) computers	✔ **b) proper behaviour**	c) numbers	d) nature

13. Identifying and obeying the principles of being online is called:

a) media bias	✔ **b) digital ethics**	c) piracy	d) phishing

14. Which is a digital ethical issue?

a) 5G	✔ **b) Deep fakes**	c) Robots	d) Smart devices

15. A fake warning saying "System may have detected viruses" is a sign of:

a) phishing	✔ **b) tech support scam**	c) copyright	d) CRM

16. According to the U.S. Census Bureau, e-commerce was __ of all retail sales in 2020.

a) 4 percent	✔ **b) 14 percent**	c) 40 percent	d) 24 percent

17. Facebook Pixel is a tool used in:

✔ **a) marketing**	b) gaming	c) piracy	d) sleeping

18. Which is a document storage and collaboration tool?

a) Spotify	✔ **b) Google Drive**	c) Miniclip	d) Pinterest

19. Copying a whole text is called __ plagiarism.

a) patchwork | ✔ **b) global** | c) self | d) paraphrasing

20. Recycling your own old work is called __ plagiarism.

a) global | b) patchwork | ✔ **c) self** | d) paraphrasing

21. Copying phrases from different sources and joining them is __ plagiarism.

✔ **a) patchwork** | b) global | c) self | d) spin

22. Piracy is a/an:

a) good habit | ✔ **b) criminal act** | c) type of bias | d) social network

23. Media bias occurs when a news outlet allows __ to affect the news.

✔ **a) opinions** | b) computers | c) advertisements | d) weather

24. Which is NOT a type of media bias?

a) Spin bias | b) Omission bias | c) Slant bias | ✔ **d) Copyright bias**

25. Violating software licensing or copyright is called:

a) unauthorized access | b) improper use | ✔ **c) illegal use** | d) phishing

26. A person's right to control how his personal information is used is called:

a) confidentiality | ✔ **b) privacy** | c) piracy | d) bias

27. VPN stands for:

a) Very Private Network | ✔ **b) Virtual Private Network** | c) Virtual Public Network | d) Visual Private Network

28. Which mode deletes cookies and temporary files after browsing?

✔ **a) Incognito mode** | b) Sleep mode | c) Safe mode | d) Full screen mode

29. Trojan horses and worms can be stopped by updated:

a) browser | ✔ **b) anti-malware** | c) VPN | d) password

30. Bullying someone using electronic means is called:

a) FOMO | ✔ **b) cyberbullying** | c) phishing | d) piracy

31. FOMO stands for:

✔ **a) Fear of Missing Out** | b) Fear of Moving On | c) Fun of Missing Out | d) Fear of Media Online

32. Spending too much time on social media can cause:

a) better sleep | ✔ **b) unhealthy sleep patterns** | c) more money | d) fast computer

Mistakes Found in the Book

1. Page 110 answer key, Unit 5 MCQ 9: The key gives (b) plagiarism. But the book (page 87) defines illegally reproducing copyrighted material as **piracy**. So the correct answer is **(c)**.
2. Page 93, MCQ 2: The options are lettered a, c, b, c (two options are "c"). The correct answer is the first one, **"Slow down your computer"**.
3. Page 88: The book writes "SAO (Service Oriented Architecture)". The correct short form is **SOA**. The same line also says "(that's piracy)", which is not correct; interfering with others is **harassment**, not piracy.
4. Page 91, Figure 5.16: The sleep stages picture shows "Stage 4" two times. The stages should be numbered **1, 2, 3 and 4**.
5. Page 85: The book uses the full form "Client Relationship Management". Most books also call CRM "Customer Relationship Management". For this book's exam, write **Client Relationship Management**.

Computer Science — Class 7

Single National Curriculum (SNC) — Punjab Curriculum and Textbook Board

Unit 6 — Entrepreneurship in Digital Age

TOPIC 1: BUSINESS PLAN: A business plan is one of the most important steps to run a successful business, whether it is run by one person (sole trader), a partnership or a limited company.

Define business plan and its components (6.1.1)

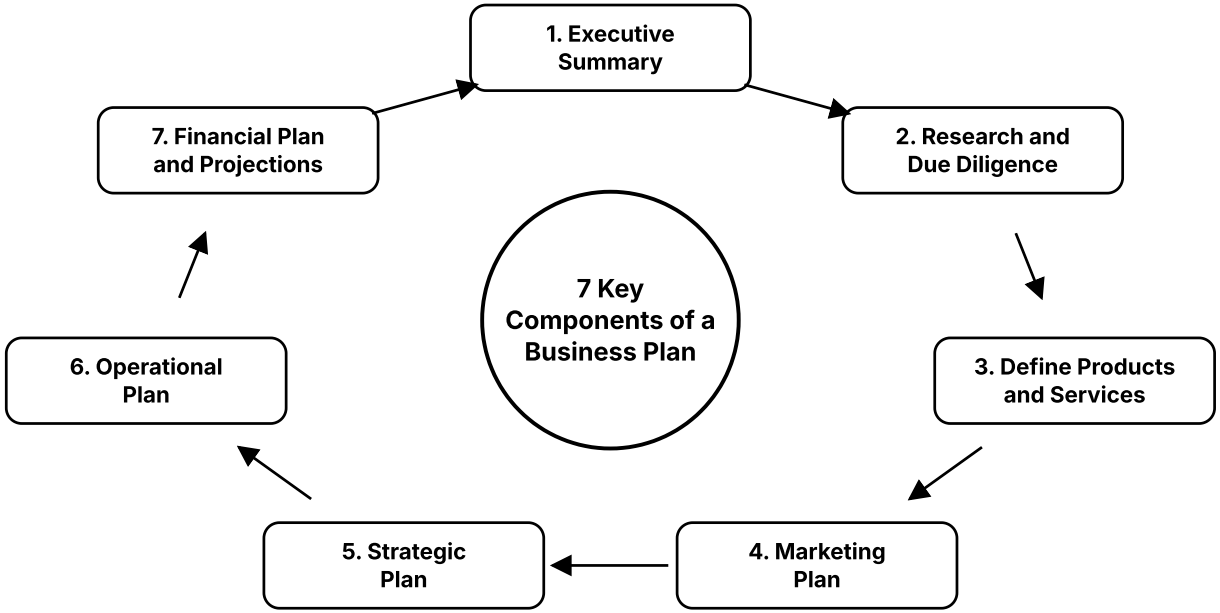
A business plan is an essential **written document**. It gives a **detailed description** and a **complete overview of the company's future**.

1. All businesses must have a business plan.
2. It explains the **business strategy** and the **key goals**: where the business is now and where it wants to be in the future.
3. It is also an important tool for an old (established) company that is moving in a **new direction**.
4. It can act as a **benchmark** (standard) to check the performance of the company.

The main areas of a business plan are goals, marketing, competitors, finance, strategy, timeline, operations and team.

Key sections of a business plan

1. An executive summary
2. A business description
3. Details of market strategies
4. Competitor analysis
5. A design and development plan of your products and services
6. Information about your operations and management plan
7. Financial information, planning and factors



7 key components of a business plan

Long Question: Write a note on key components of a business plan. *[Exercise (Brief Q8)]*

1. **Executive Summary:** It gives a short overview of the whole plan. It may include a **table of contents**, company background, market opportunity, management overview, competitive advantages and **financial highlights**.
2. **Business Description and Structure:** It explains **why** you are in business and **what products** you sell. It also describes the manufacturing process, availability of materials, how you handle stock (inventory) and delivery, and other operational details. For services, it describes them and their value to customers. It also includes strategic relationships, office (administrative) issues, intellectual property, expenses and the **legal structure** of the company.
3. **Competitor Analysis:** Before starting a business in a market, we must **research the competitors**. Complete research helps us to make suitable plans to meet the demand and supply of the product or service.
4. **Development Plan of Products:** It is the path of **how the plan will be carried out**. The product or service must be made by following a set of instructions, with **quality control** and careful checks.
5. **Market Research and Strategies:** It explains the market analysis and the marketing strategy, including **sales forecasts, deadlines and milestones, advertising and public relations**. This is how we compete with others. If we do not have much data, we can give **testimonials** (good comments) from existing customers.

6. **Management and Personnel:** It gives short life histories (bios) of the company's executives and managers, and explains how their skills will help to meet business goals. Investors check the **risk**, and an **experienced** management team can lower the risk.
7. **Financial Documents:** It gives the **numbers** that support everything in the plan. It includes careful projections of the **profit and loss statement, balance sheet and cash flow statement** for the **next three years**.

Long Question: What is the use of a business plan in the ideology of developing a business? ***[Exercise (Brief Q9)]***

1. **Clear picture:** A business plan is a written document that gives a complete overview of the **company's future**. It tells what the business will do and how.
2. **Goals and strategy:** It explains the business strategy and the key goals, from where the business is now to where it wants to be.
3. **Guide for work:** It works like a **road map**. The development plan tells how the product will be made, and the operations plan tells how the business will run.
4. **Getting money:** Investors read the business plan and the financial documents to decide whether to put money in the business.
5. **New direction:** It is also useful for an old company that wants to move in a new direction.
6. **Benchmark:** It acts as a **benchmark** to check whether the company is performing well.

Long Question: Why is market research important when planning to set up a business? ***[Exercise (Brief Q10)]***

1. **Knowing the competitors:** Before starting a business, research must be done on the **competitors**. It tells us what others are selling, at what price and how.
2. **Demand and supply:** Complete research helps us to make suitable strategies to meet the **demand and supply** of the product or service.
3. **Marketing strategy:** Market research helps us to plan the **sales forecast**, deadlines, milestones, advertising and public relations.
4. **Beating the competition:** This is how we **combat our competition** and make a successful business plan.
5. **Building trust:** If we cannot produce a lot of data, we can show **testimonials** from existing customers.

Conclusion: Without market research, a business may make a product that people do not need, or sell it at the wrong price. So market research reduces risk.

TOPIC 2: PROMOTION, VALUE PROPOSITION AND QUALITY ASSURANCE: These three ideas help a company to sell its products and keep its customers happy.

Promotion (6.2)

Promotion is the **whole set of activities** that tell the user about a product, brand or service. The aim is to make people **aware**, **attract** them and **persuade** them to buy the product instead of others.

1. **Types:** Advertisements, press releases and consumer promotions (schemes, discounts and contests).
2. **Below the line:** Trade discounts, freebies, incentive trips, awards and so on.

3. Sales promotion is a part of the overall promotion effort.

Value proposition (6.3)

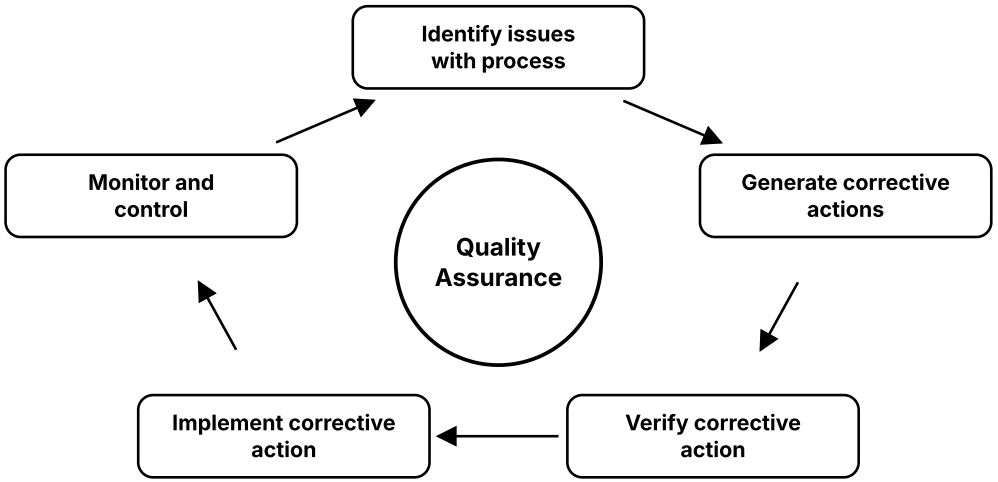
A value proposition in marketing is a **short (concise) statement of the benefits** that a company gives to the customers who buy its products or services. It is a declaration of intent, both inside the company and in the market.

Do you know? The term value proposition first appeared in a **McKinsey & Co.** research paper in **1988**. It defined it as "a clear, simple statement of the benefits, both tangible and intangible, that the company will provide, along with the approximate price it will charge each customer segment for those benefits."

Quality assurance (6.4)

Quality assurance (QA) is a process that helps a business follow the **quality standards** of its product, set by the company or its industry. It is the company's process for **improving the quality** of its products.

1. **Promise:** Many businesses see QA as a promise to their stakeholders and customers that they will deliver **high-quality products** with a good user experience.
2. **Broad process:** QA **prevents** quality failures. The QA team works in **all stages** of a product: production, testing, packaging and delivery.
3. **Quality control:** It is a **narrower** process. It focuses on **finding mistakes, errors or missing requirements** in a product.



The quality assurance cycle

TOPIC 3: PROJECT MANAGEMENT AND MEDIA LITERACY: Project management helps a business finish its work on time. Media literacy helps people understand media messages.

Importance of project management and media literacy (6.5)

Handling projects can be a difficult task. It needs a good understanding of project **scheduling, planning, reporting and tracking**. A good project manager must understand project management and its job roles.

1. **Project Management:** The Project Management Institute says, "Project management is the application of **skills, knowledge, tools and techniques** to project activities to meet project requirements." It also uses policies, principles and procedures to guide a project from the **start until it is complete**. It gives **leadership and direction** to projects.
2. **Media Literacy:** Media literacy helps people to **understand** and take part in public discussion, and finally to make **sound (good) decisions** when choosing (electing) their leaders. Media literate people can **think critically** about media messages.

Long Question: How does project management help in developing a business plan? ***[Exercise (Brief Q7)]***

1. **Meaning:** Project management is the use of skills, knowledge, tools and techniques to complete project activities and meet the project's requirements.
2. **Planning and scheduling:** A business plan has many tasks, like market research, product development and finance. Project management **plans and schedules** these tasks.
3. **Reporting and tracking:** It **reports** and **tracks** the progress, so that deadlines and milestones in the business plan are met.
4. **From start to finish:** It uses policies, principles and procedures to guide the work from the **first stage until completion**.
5. **Leadership:** It gives **leadership and direction** to the team, so everybody knows their job.
6. **Media literacy:** Media literacy also helps, because the business can understand media messages and public opinion while planning its marketing.

TOPIC 4: PAYMENTS AND TRANSACTIONS: Payments and transactions are part of every business. They are related, but they are not the same.

Payments (6.6)

Payment is the **voluntary transfer** of money, or something of equal value, from one person to another in exchange for goods or services, or to meet a **legal obligation**.

1. **Payer:** The person who **gives** the money.
2. **Payee:** The person who **gets** the money.
3. **Ways to pay:** Cash, cheque, wire transfer, credit card, debit card or even crypto currency.

Transactions (6.7)

A transaction is a **complete agreement** between a **buyer and a seller** to exchange goods, services or financial assets for money. The term is also used in company (corporate) accounting.

Long Question: Elaborate the difference between a transaction and a payment. ***[Exercise (Brief Q1)]***

Payment	Transaction
It is the transfer of value (money or goods) from one party to another.	It is a complete agreement between a buyer and a seller.
It is given for goods or services, or to meet a legal obligation.	It is made to exchange goods, services or financial assets for money.
It can take many forms: cash, cheque, card, wire transfer, crypto currency.	It is one instance of buying or selling something.
Barter (exchanging one good or service for another) is also a form of payment.	In accounting, events that affect a business's money are recorded as accounting transactions.
It involves a payer (who gives) and a payee (who gets).	It involves a buyer and a seller.

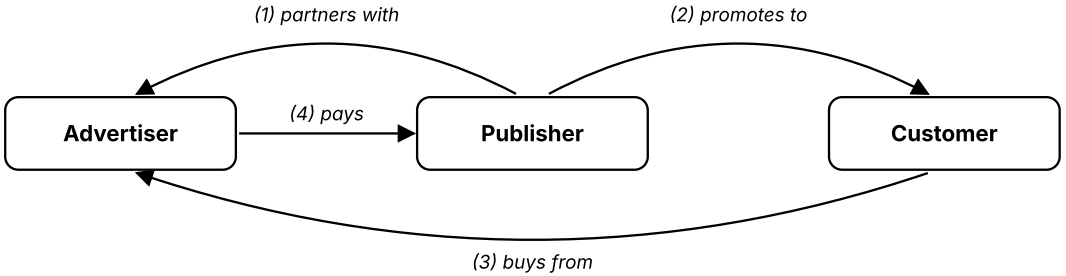
Example: When you buy a book from a shop, the whole deal is a **transaction**, and the money you give to the shopkeeper is the **payment**.

TOPIC 5: DIGITAL MARKETING: Technology has become very popular and important in business in the last ten years. Digital marketing helps companies to improve their sales and brand image.

Long Question: Write a note on the tools and techniques used in digital marketing. ***[Exercise (Brief Q2)]***

There are **9** different digital marketing techniques:

1. **Social Media Marketing:** A strong presence on social media platforms is the **most important** digital marketing tool. It is an essential technique to grow digitally.
2. **Search Engine Optimization (SEO):** It brings **more visitors (traffic)** to a website by making it appear **higher** in the results of a search engine like Google. It improves brand awareness.
3. **Email Marketing:** It brings customers **back** to a brand to buy new products. People sign up for an email list after a purchase. This builds customer loyalty and relationship.
4. **Content Marketing:** It means creating **valuable and useful content** regularly on platforms. It improves brand awareness.
5. **Video Marketing:** It works with content marketing. With good-quality commercials, companies leave a **long-lasting brand image** in people's minds.
6. **Web Advertising:** Brands make **clickable advertisements** and put them on popular websites.
7. **Affiliate Advertising:** A brand **pays** a famous person with a large audience, usually a **blogger or social media influencer**, to post about the company.
8. **Overall Personalization:** Giving the brand a **unique style** makes people believe the company is successful.
9. **Creating an App:** An app gives customers an easy way to contact the brand, see **where to buy**, and get **notified about new products**.



How affiliate marketing works

Long Question: What role does digital marketing play in today's world? ***[Exercise (Brief Q3)]***

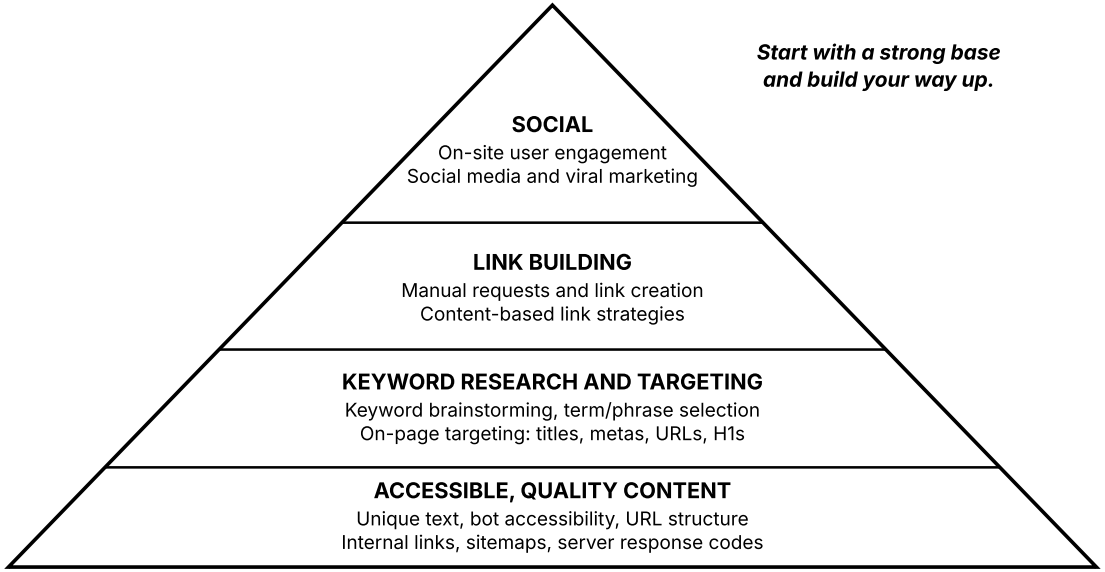
1. **Technology in business:** In the last ten years, technology has become very popular and important in the business world.
2. **More sales:** Digital marketing helps companies to **improve their sales** through social media, websites, emails and apps.
3. **Brand image:** It helps to build a strong **brand image** and brand awareness, for example through content marketing and video marketing.
4. **Reaching more people:** With SEO, web advertising and affiliate advertising, a business can reach a **very large audience** at low cost.
5. **Keeping customers:** Email marketing and apps bring customers **back** and tell them about new products. This builds **loyalty**.
6. **Two-way communication:** Customers can talk to the brand easily, and the brand can treat each customer in a personal way.

TOPIC 6: SEARCH ENGINE OPTIMIZATION AND SOCIAL MEDIA:

SEO stands for **Search Engine Optimization**. It is the process of improving a website so that it can be seen more easily in search results.

Search engine optimization (6.9)

SEO increases the **visibility** of a website when people search for products or services in **Google, Bing** and other search engines. The better the visibility, the more likely we are to attract new and existing customers.



The SEO pyramid: start with a strong base and build up

Long Question: How does the search engine optimization help the business grow? ***[Exercise (Brief Q4)]***

1. **Better visibility:** SEO improves a website so that it appears **higher** in the search results of Google, Bing and other search engines.
2. **More customers:** The better the visibility, the more **attention** the business gets. It attracts **new (prospective) and existing customers**.
3. **Huge number of searches:** People make **trillions of searches** every year, often to find information about products and services they want to buy.
4. **Main source of traffic:** Search is often the **main source of digital traffic** for brands, and it supports other marketing channels.
5. **Beating competitors:** Ranking higher than competitors in search results has a big effect on the **profit (bottom line)** of the business.
6. **Brand awareness:** SEO also improves the brand's awareness.

Twitter (6.9.1)

1. Twitter came from a podcasting company called **Odeo**, started in **2004** by **Evan Williams, Biz Stone and Noah Glass**.
2. In 2005, Apple added podcasts to **iTunes**, so Odeo needed a new direction.
3. Engineer **Jack Dorsey** suggested a **short message service (SMS)** to share small blog-like updates with friends. Glass suggested the name **"Twttr"**.
4. Dorsey sent the **first tweet** ("just setting up my twttr") on **March 21, 2006**. The complete Twitter started in **July 2006**.
5. In October 2006, Williams, Stone and Dorsey bought Odeo and started **Obvious Corp.**

Snapchat (6.9.2)

1. **Evan Spiegel** and **Bobby Murphy** met at **Stanford University**. In spring **2011**, they made **Picaboo (Peekaboo)**, the first version of Snapchat.
2. It was an app to send photos that **disappear** after some time. It was launched in the iOS App Store in July.
3. Users could take **screenshots**, which made disappearing photos useless. So they added a **notification** that tells when someone takes a screenshot.
4. In September, they renamed (rebranded) it **Snapchat**, added **captions** to photos and launched it again. They focused on technology more than on advertising.

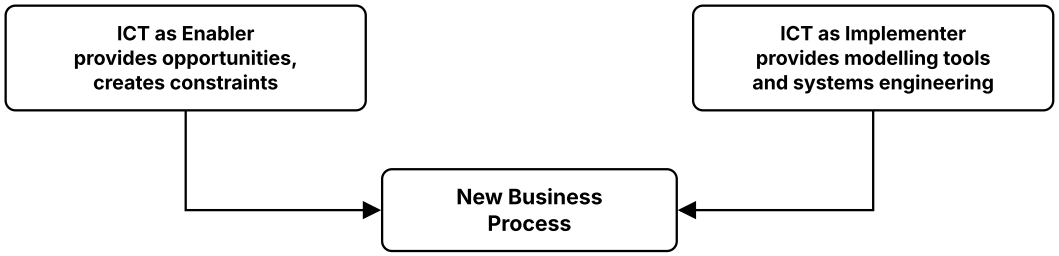
Blogs (6.9.3)

A blog (short for **"weblog"**) is an **online journal** or information website. It shows posts in **reverse chronological order**, so the **latest post appears first**, at the top. A writer or a group of writers share their views on a subject with people all over the world.

TOPIC 7: TECHNOLOGY AND DIGITAL PLATFORMS FOR ENTREPRENEURSHIP: Technology plays a very important role in today's modern world in helping entrepreneurship to grow.

Long Question: How does the technology promote the entrepreneurship in the economy? ***[Exercise (Brief Q5)]***

1. **Enablers:** Entrepreneurship enablers are those things that make it **possible for entrepreneurs to start and grow**. Technology is one of the biggest enablers.
2. **Infrastructure and culture:** Technology improves the **infrastructure** (like the Internet and payment systems) and the business culture.
3. **ICT as an enabler:** ICT gives **opportunities** (and also some limits) to entrepreneurs.
4. **ICT as an implementer:** ICT gives **modelling tools and systems engineering** to build new business processes.
5. **Digital entrepreneurship:** Old businesses are becoming digital, and new **digital enterprises** are being created. This brings innovation and **economic growth**.
6. **Digital platforms:** Platforms like Facebook, Instagram, Amazon and PayPal let people sell, market and receive payments easily.



ICT as enabler and implementer creates a new business process

Digital platforms used for entrepreneurship (6.11)

Entrepreneurship is a main driving force of **innovation and economic growth**. Digital entrepreneurship is rising, through the **digitization of existing businesses** and the creation of **digital enterprises**.

A **Digital Business Platform** provides a way to **recombine technologies** to deliver new business capabilities in new and creative ways. It gives a logical layer that separates the business logic from the technology applications behind it.

Long Question: State and explain the types of digital platforms used for entrepreneurship. ***[Exercise (Brief Q6)]***

Platforms in digital business exist at many levels. They can be high-level platforms that support a whole platform business model, or low-level platforms that give business or technology capabilities to other products. Common digital platforms are:

1. **Social media and media-sharing platforms:** They let people share posts, photos and videos. *Examples:* Instagram, YouTube, Facebook, Twitter and LinkedIn.
2. **Knowledge platforms:** People ask questions and share knowledge. *Examples:* Quora and Reddit.
3. **Service-oriented platforms:** They connect people who give a service with people who need it. *Example:* Airbnb.
4. **E-commerce platforms:** They are used to buy and sell goods online. *Example:* Alibaba.

Types of platform business models

Platform type	Examples
Social Media	Facebook, LinkedIn, Twitter
Communication	WhatsApp, Skype, WeChat, LINE
Search	Google, Bing, Yahoo, Safari
Sharing Economy	Airbnb, Uber (peer-to-peer), ZipCar; file and music sharing
Matching	TaskRabbit
Development	App stores, gaming consoles
Content and Review	YouTube, TripAdvisor, Yelp
Booking Aggregators	Booking.com, Expedia, Agoda
Retail	Amazon, Etsy, eBay, Craigslist
Payment	PayPal, Alipay, Visa
Crowdsourcing and Crowdfunding	InnoCentive, Kickstarter

SHORT QUESTIONS — UNIT 6

1. Elaborate the difference between a transaction and a payment. ***[Exercise]***

A payment is the transfer of money or value from a payer to a payee. A transaction is the complete agreement between a buyer and a seller to exchange goods or services for money.

2. Write the names of tools and techniques used in digital marketing. ***[Exercise]***

Social media marketing, SEO, email marketing, content marketing, video marketing, web advertising, affiliate advertising, overall personalization and creating an app.

3. What role does digital marketing play in today's world? ***[Exercise]***

It improves sales and brand image, reaches many people at low cost, and brings customers back through emails and apps.

4. How does search engine optimization help the business grow? ***[Exercise]***

SEO makes a website appear higher in search results, so more customers see it and visit it. This increases sales and profit.

5. How does technology promote entrepreneurship? ***[Exercise]***

Technology makes it possible for entrepreneurs to start and grow. It gives opportunities, tools and digital platforms to sell and get paid.

6. State the types of digital platforms. ***[Exercise]***

Social media platforms (Instagram, YouTube), knowledge platforms (Quora, Reddit), service-oriented platforms (Airbnb) and e-commerce platforms (Alibaba).

7. How does project management help in developing a business plan? ***[Exercise]***

It plans, schedules, reports and tracks the tasks of the plan and guides the work from start to finish.

8. What is the use of a business plan? ***[Exercise]***

It explains the business strategy and goals, guides the work, helps get money from investors and acts as a benchmark for performance.

9. Why is market research important when setting up a business? ***[Exercise]***

It tells us about competitors and demand and supply, so we can make a good marketing strategy and beat the competition.

10. Define business plan. ***[Book concept]***

A business plan is an essential written document that gives a detailed description and complete overview of the company's future.

11. Name the seven key sections of a business plan. ***[Book concept]***

Executive summary, business description, market strategies, competitor analysis, product development plan, operations and management plan, and financial information.

12. What is included in an executive summary? ***[Book concept]***

Table of contents, company background, market opportunity, management overview, competitive advantages and financial highlights.

13. What is promotion? ***[Book concept]***

Promotion is the whole set of activities that tell the user about a product, brand or service and persuade them to buy it.

14. What is a value proposition? ***[Book concept]***

It is a short statement of the benefits that a company gives to the customers who buy its products or services.

15. Differentiate between quality assurance and quality control. ***[Book concept]***

Quality assurance is a broad process that prevents quality failures in all stages. Quality control is narrower and finds mistakes and errors in a product.

16. Define project management. ***[Book concept]***

It is the application of skills, knowledge, tools and techniques to project activities to meet project requirements.

17. What is media literacy? ***[Book concept]***

Media literacy helps people understand media messages, take part in public discussion and make good decisions when electing leaders.

18. Who are payer and payee? ***[Book concept]***

The payer is the person who gives the money. The payee is the person who gets the money.

19. What is affiliate advertising? ***[Book concept]***

A brand pays a blogger or social media influencer with a large audience to post about the company.

20. What is a blog? ***[Book concept]***

A blog (weblog) is an online journal that shows the latest post first, at the top.

21. Who sent the first tweet and when? ***[Book concept]***

Jack Dorsey sent the first tweet, "just setting up my twttr", on March 21, 2006.

22. What was the first name of Snapchat? ***[Book concept]***

Picaboo (Peekaboo), made by Evan Spiegel and Bobby Murphy in 2011.

23. What is a digital business platform? ***[Book concept]***

It is a way to recombine technologies to deliver new business capabilities in new and creative ways.

MULTIPLE CHOICE QUESTIONS — UNIT 6

1. A __ is an essential written document that provides a detailed description of company's future. ***[Exercise MCQ 1]***

a) Quality assurance	b) Entrepreneurship	✔ **c) Business plan**	d) Project management

2. __ may include a table of content, company's background, market opportunity and financial highlights. ***[Exercise MCQ 2]***

a) Business description	b) Competitors analysis	c) Financial documents	✔ **d) Executive summary**

3. __ can be defined as a set of activities which communicates the product, brand or services to the user. ***[Exercise MCQ 3]***

a) Value proposition	✔ **b) Promotion**	c) Quality assurance	d) Entrepreneurship

4. A __ is a completed agreement between a buyer and seller to exchange goods, services or financial assets for money. ***[Exercise MCQ 4]***

a) Payment	✔ **b) Transaction**	c) Financial statements	d) Contract

5. __ is the digital marketing technique that involves creating more traffic to a website by making it appear higher in search results. ***[Exercise MCQ 5]***

✔ **a) Search engine optimization**	b) Email marketing	c) Social media marketing	d) Content marketing

6. A __ is an online journal or information website displaying information. ***[Exercise MCQ 6]***

a) Tweet	b) Snapchat	✔ **c) Blogs**	d) Facebook profile

7. A __ in marketing is a concise statement of the benefits that a company is delivering to customers. ***[Exercise MCQ 7]***

			✔
a) Competitors analysis	b) Quality assurance	c) Promotion	**d) Value proposition**

8. __ includes the sales forecast, deadlines and milestones, advertising and public relations. ***[Exercise MCQ 8]***

		✔	
a) Financial document	b) Management and personnel	**c) Market research and strategies**	d) Development plan of your product

9. __ enables the populace to understand and contribute to public discourse and make sound decisions when electing their leaders. ***[Exercise MCQ 9]***

	✔		
a) Project management	**b) Media literacy**	c) Quality assurance	d) Social media marketing

10. __ is the voluntary transfer of money or other valuable items from one person to another in exchange of goods and items. ***[Exercise MCQ 10]***

✔			
a) Payments	b) Transactions	c) Barter	d) Contracts

11. How many key sections does a business plan have?

		✔	
a) five	b) six	**c) seven**	d) nine

12. Profit and loss statement, balance sheet and cash flow statement are part of:

	✔		
a) executive summary	**b) financial documents**	c) competitor analysis	d) promotion

13. Financial projections in a business plan are made for the next:

	✔		
a) one year	**b) three years**	c) ten years	d) six months

14. Trade discounts, freebies and awards are __ promotions.

✔			
a) below the line	b) above the line	c) online	d) value

15. The term value proposition first appeared in a McKinsey & Co. paper in:

	✔		
a) 1978	**b) 1988**	c) 1998	d) 2008

16. Finding mistakes and errors in a product is called:

	✔		
a) quality assurance	**b) quality control**	c) promotion	d) marketing

17. The person who gets the money is called the:

	✔		
a) payer	**b) payee**	c) buyer	d) banker

18. Exchanging one good or service for another is called:

✔			
a) barter	b) transaction	c) promotion	d) credit

19. How many digital marketing techniques are given in the book?

		✔	
a) five	b) seven	**c) nine**	d) ten

20. The most important digital marketing tool is:

✔			
a) social media marketing	b) web advertising	c) creating an app	d) email marketing

21. A brand paying an influencer to post about it is called:

	✔		
a) content marketing	**b) affiliate advertising**	c) SEO	d) email marketing

22. SEO stands for:

✔			
a) Search Engine Optimization	b) Search Email Operation	c) Social Engine Optimization	d) Search Engine Organization

23. The base of the SEO pyramid is:

		✔	
a) social	b) link building	**c) accessible, quality content**	d) keyword research

24. Twitter came from the podcasting company:

a) Apple	✔ **b) Odeo**	c) Obvious	d) iTunes

25. The first tweet was sent by:

a) Evan Williams	✔ **b) Jack Dorsey**	c) Biz Stone	d) Noah Glass

26. The first version of Snapchat was called:

a) Twttr	✔ **b) Picaboo**	c) Odeo	d) Weblog

27. A blog shows the latest post:

a) at the bottom	✔ **b) at the top**	c) in the middle	d) nowhere

28. Quora and Reddit are __ platforms.

a) e-commerce	✔ **b) knowledge**	c) payment	d) booking

29. Airbnb is a __ platform.

a) knowledge	✔ **b) service-oriented**	c) search	d) payment

30. Alibaba is an example of a/an __ platform.

a) knowledge	✔ **b) e-commerce**	c) communication	d) search

31. PayPal and Alipay are __ platforms.

a) retail	b) search	✔ **c) payment**	d) matching

Mistakes Found in the Book

1. Page 110 answer key, Unit 6: The key counts the options in the order they are **printed** (a, c, b, d), not by their letters. So four answers in the key are wrong by letter: **MCQ 1** is (c) Business plan, not (b); **MCQ 3** is (b) Promotion, not (c); **MCQ 8** is (c) Market research and strategies, not (b); **MCQ 9** is (b) Media literacy, not (c). These notes tick the correct options.
2. Unit 6 exercise has no "Answer in detail" section. The 10 brief questions are the only exercise questions, so all of them are solved as long questions, and also given in short form.
3. Page 102: "Evan William" should be "Evan **Williams**". Page 103 calls the first Snapchat app both "Peekaboo" and "Picaboo"; its real name was **Picaboo**.
4. Page 105: The platform chart writes "Pagoda" for **Agoda**, and lists "Safari" (a web browser) as a search platform.